



5

# Electrical currents

5.1

## Electric potential difference, current and resistance

### Assessment statements

- 5.1.1 Define electric potential difference.
- 5.1.2 Determine the change in potential energy when a charge moves between two points at different potentials.
- 5.1.3 Define the electronvolt.
- 5.1.4 Solve problems involving electric potential difference.
- 5.1.5 Define electric current.
- 5.1.6 Define resistance.
- 5.1.7 Apply the equation for resistance in the form  $R = \frac{\rho l}{A}$  where  $\rho$  is the resistivity of the material of the resistor.
- 5.1.8 State Ohm's law.
- 5.1.9 Compare ohmic and non-ohmic behaviour.
- 5.1.10 Derive and apply expressions for electrical power dissipation in resistors.
- 5.1.11 Solve problems involving potential difference, current and resistance.



All electrical appliances convert electrical energy into other forms.

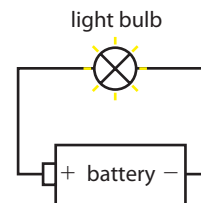
In this chapter we will develop the theory behind electric circuits, but first we need to understand how the individual components work.

### The battery

All electrical devices convert electrical energy into other forms; an MP3 player produces sound, a torch produces light and an electric motor produces mechanical energy. In cordless devices (ones that don't plug into the mains) the source of this energy is the battery.

Inside the battery, chemicals react with each other to convert chemical energy into electrical energy.

When the light bulb is connected across the ends of the battery, it gets hot and starts to glow, giving out energy in the form of light and heat. This energy comes from the battery. To make the bulb light up, we must make a complete circuit. If this circuit is broken, the bulb goes out. It seems like something is flowing through the wire; we call the flow *current* and the thing that is flowing we call *charge*.



**Figure 5.1** This simple circuit consists of a battery and a light bulb. The battery has two ends, one is positive (+) and the other is negative (–). To make a circuit, the bulb must be connected as shown.



To build your own circuits and test the theories in this chapter, visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on the Weblink 5.1.

**Figure 5.2** Water flow is similar to the flow of electricity.

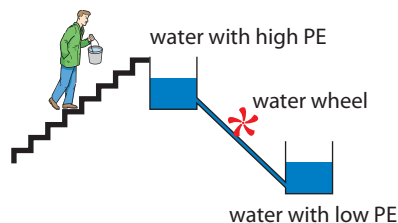
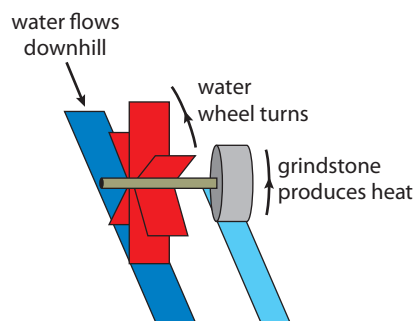


Figure 5.2 shows a man carrying water up the stairs and pouring it into a tank. The water flows from this tank through a pipe into a lower one. As it flows, the water turns a water wheel. The water wheel turns a grindstone that rubs against a hard surface, producing heat.

### Energy explanation

We can explain what is happening using the principle of conservation of energy. As the man carries the water up the stairs he does work on the water, increasing its potential energy(PE). When the water flows downhill into the lower tank, it loses PE. This energy is given to the water wheel, which turns the grindstone, producing heat. So we can say that the work done by the man is the same as the heat produced at the grindstone.

**Figure 5.3** Energy changes.



### Charge

Unlike with water, we cannot see charge flowing through the wire. We will find out later (Chapter 6) that there are two types of charge (+ and -) but to make things simple we will just consider + charge. The unit of charge is the coulomb (C).

### Electrical potential energy

To make the water flow through the pipe, the man did work on the water to raise its potential energy. In the electrical circuit, the battery does work to increase the electrical potential energy of the charges. The charges on the + end of the battery have a higher potential energy than the ones on the - end, so charge flows from + to - outside the cell.

### Current

Current is the flow of charge, and since charges at the + end of the battery have higher PE than those at the - end, the current flows from + to -.

The unit of current is the amp (A).

The amp is a scalar quantity.

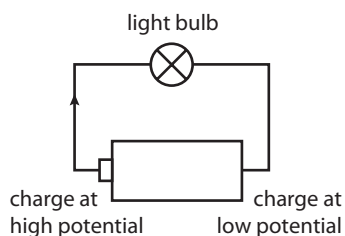
$$\text{Current} = \frac{\text{charge}}{\text{time}}$$

1 amp = 1 coulomb per second.



## The complete picture

The battery takes charges and puts them in a position of high potential at the + end. If a bulb is connected between the ends of the battery, this charge flows from the position of high potential to a position of low potential. The potential energy is converted to heat and light.



## Microscopic model

Metals allow charge to flow because they contain small negatively charged particles called *electrons*. Since the electrons are negative, they flow from low potential to high potential. Trying to imagine something flowing uphill is not very easy, so we will stick with positive charges flowing downhill. This is not a problem but we must always remember that electrons flow in the opposite direction to conventional current.

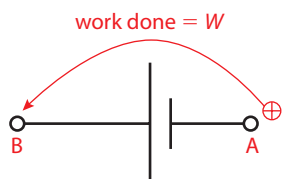
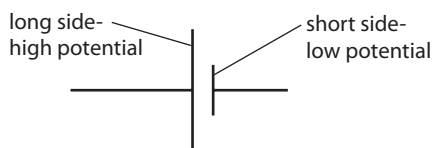
## Energy transformation in a battery

Since energy is conserved, the energy given to the charges must come from somewhere — in fact, the battery converts chemical energy to electrical PE. The chemicals in the battery contain charges that are all mixed up. When these chemicals react, the charges are rearranged so that there are more + charges at the + end of the battery. After the reaction, the chemicals have less energy, as the chemical energy has been converted to electrical PE.

## Potential difference

If we move a + charge from the negative end of the battery to the positive end, we need to do work. This is because the charge has a higher potential energy at the positive end than the negative end. The amount of work done per unit charge is defined as the potential difference (p.d.) between the plates or terminals.

The unit of p.d. is the volt. The volt is a scalar quantity. 1 volt = 1 joule per coulomb.



**Figure 5.4** As the charge flows from + to –, its potential energy is converted into heat and light in the bulb.

The electron is a small particle responsible for carrying charge in conductors.  
Charge =  $-1.6 \times 10^{-19}$  C  
Electrons are negative so flow in the opposite direction to conventional current.

We consider current to flow in the opposite direction to the way that electrons actually flow. Does it matter that our model is actually completely backwards?

The PhET simulation 'Battery voltage' shows how the charges are rearranged. The little people represent chemical energy. To view this, visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on the Weblink 5.2.

**Figure 5.5** The symbol for a cell makes a lot of sense; the side that is at the highest potential has the longest line.

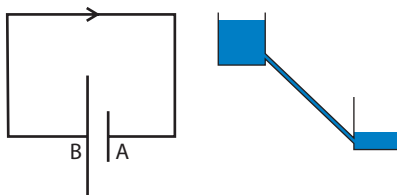
The p.d. between A and B is defined as the amount of work done per unit charge in taking a small +ve test charge from A to B.

**Figure 5.6** If an amount of work  $W$  is done in taking charge  $q$  from A to B then the p.d. ( $V$ ) between A and B is given by the equation  $V = W/q$ .

## Conductors and insulators

A *conductor* is a material that allows charge to flow through it. An *insulator* doesn't. All metals are good conductors of electricity, but plastics are insulators.

**Figure 5.7** Current flows from B to A like water flowing downhill.



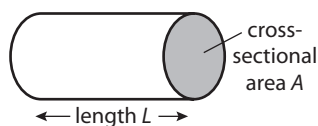
## Worked examples

- 1 If a current of 10 A flows for 15 s, how much charge flows?
- 2 How much energy is lost when +5 C of charge flows from the +ve terminal to the -ve terminal of a 9 V battery?

### Solutions

- 1 Current  $I = \text{charge per unit time} = \frac{Q}{t}$   
 $Q = It$   
 $= 10 \times 15 \text{ C}$   
 $= 150 \text{ C}$
- 2 Energy to move 1 C = 9 J  
 Energy to move 5 C =  $5 \times 9 \text{ J} = 45 \text{ J}$

● **Examiner's hint:** From definition of p.d., the energy lost per unit charge moving from one terminal to the other is  $9 \text{ J C}^{-1}$ .



**Figure 5.8** Dimensions of a conductor.

## Resistance

The rate at which charge can flow through a conductor depends on the size and material of the conductor. A conductor that does not let much current flow for a given p.d. is said to have a high resistance.

Resistance ( $R$ ) is related to: cross-sectional area ( $A$ ), length ( $L$ ) and the material.

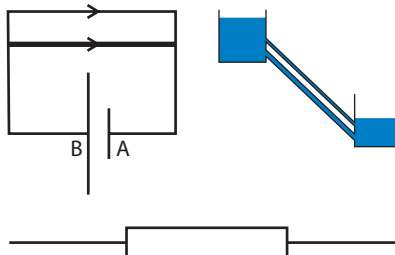
$$R \propto \frac{L}{A}$$

The constant of proportionality is called the *resistivity* ( $\rho$ ).

So 
$$R = \rho \frac{L}{A}$$

The unit of resistance is the ohm ( $\Omega$ ).

This means that more current flows through a short fat conductor than a long thin one.



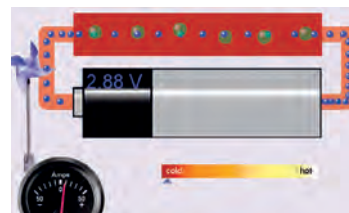
**Figure 5.9** More current flows through the bottom branch than the top one.

**Figure 5.10** The symbol for a resistor.

## Resistivity



A resistor is a component with a known resistance. You can work out the resistance from the colours.



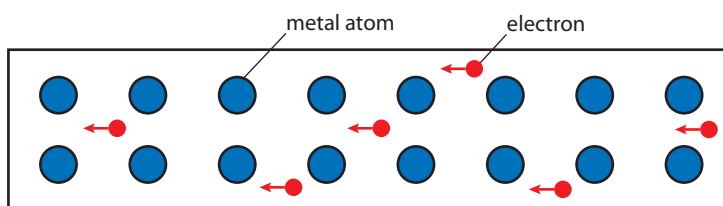
**W** To try the battery-resistance circuit, visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on Weblink 5.3.

By rearranging  $R = \rho \frac{L}{A}$  we get  $\rho = \frac{RA}{L}$  (units  $\Omega\text{m}$ ).

From this we can deduce that if the length of a sample of material is 1 m and the cross-sectional area is  $1\text{ m}^2$ , then  $\rho = R$ . You can probably imagine that the resistance of such a large piece of metal will be very small – that's why the values of resistivity are so low (e.g. for copper  $\rho = 1.72 \times 10^{-8} \Omega\text{m}$ ).

### Microscopic model of resistance

As mentioned earlier, it is actually the negative charges called *free electrons* that move. As they move through the metal, they give energy to the metal atoms, causing the temperature of the metal to increase.



**Figure 5.11** Electrons move through metal, giving energy to atoms.

### Worked example

The resistivity of copper is  $1.72 \times 10^{-8} \Omega\text{m}$ . What is the resistance of a 1 m length of 2 mm diameter copper wire?

#### Solution

$$\begin{aligned}\text{Cross-sectional area} &= \pi (0.001)^2 \\ &= 3.14 \times 10^{-6} \text{ m}^2\end{aligned}$$

$$\begin{aligned}R &= \rho \frac{L}{A} \\ &= 1.72 \times 10^{-8} \times \frac{1}{3.14 \times 10^{-6}} \Omega \\ &= 0.0055 \Omega\end{aligned}$$

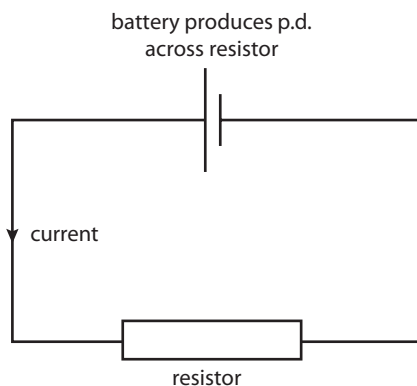
#### Examiner's hint:

$$\begin{aligned}\text{Area} &= \pi r^2 \text{ where radius} \\ &= \frac{1}{2} \times 2 \text{ mm} = 0.001 \text{ m.}\end{aligned}$$

## Ohm's law

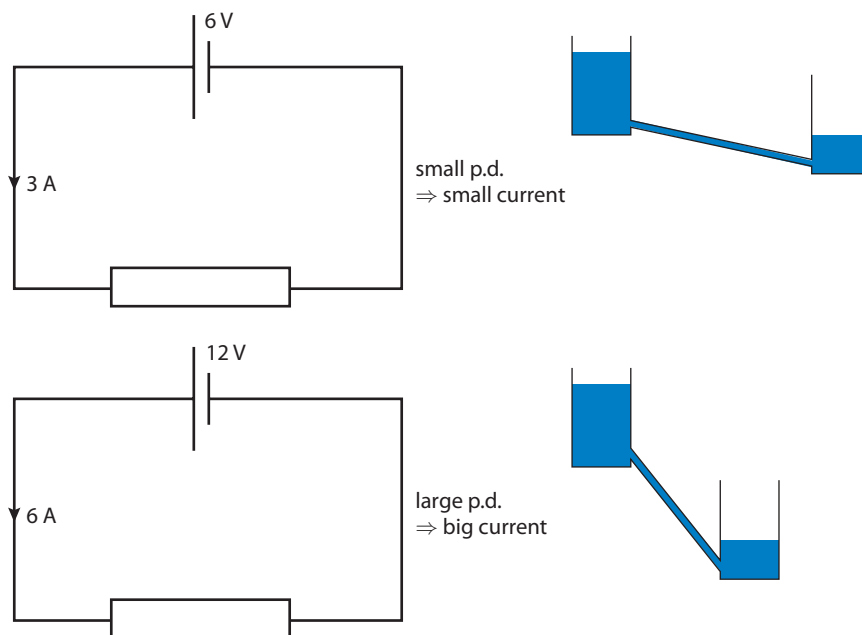
Ohm's law relates the current flowing through a conductor with the potential difference across it.

**Figure 5.12** We can now make a simple circuit with a battery and a resistor.

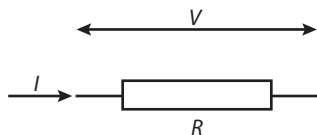


The current flowing through an ohmic conductor is directly proportional to the potential difference across it, provided temperature and other physical conditions remain constant.

**Figure 5.13** Doubling p.d. doubles the current.



**Figure 5.14**  $V$ ,  $I$  and  $R$



If the p.d. across a conductor is  $V$  and the current flowing through it is  $I$ , then according to Ohm's law:

$$V \propto I$$

The constant of proportionality is the resistance,  $R$ .

So:

$$V = IR$$

## Worked example

If the p.d. across a  $3\ \Omega$  resistance is  $9\text{ V}$  what current will flow?

### Solution

$$V = IR \quad \text{From Ohm's law.}$$

$$I = \frac{V}{R} \quad \text{Rearranging.}$$

$$I = \frac{9}{3}\text{ A}$$

$$= 3\text{ A}$$

The water behind this dam has a high potential energy but because the hole is small only a small current of water can flow out. This is equivalent to a high resistance leading to a small current.

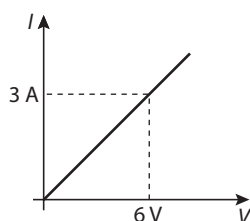


## Graphical treatment

### Ohmic conductor

Since  $V \propto I$  for an ohmic conductor, a graph of  $I$  against  $V$  will be a straight line.

In this example, resistance  $= \frac{V}{I} = 2\ \Omega$



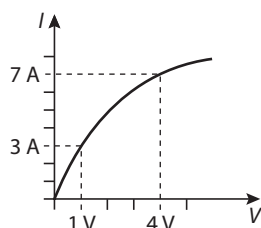
**Figure 5.15**  $I$ - $V$  for an ohmic conductor



Note: The resistance is found by taking the ratio  $\frac{V}{I}$  – this is the same as  $\frac{1}{\text{gradient}}$  for an ohmic conductor.

### Non-ohmic conductors

Not all conductors obey Ohm's law.  $I$ - $V$  graphs for these conductors will not be straight. A light bulb filament is an example of a non-ohmic conductor.



In this example, the resistance at the start is  $\frac{1}{3}\ \Omega$  ( $0.33\ \Omega$ ) and at the end it is  $\frac{4}{7}\ \Omega$  ( $0.57\ \Omega$ ).

The reason for this is that when the light bulb gets hot, the metal atoms vibrate more. This means that there are more collisions between the electrons and metal atoms, leading to an increase in resistance.

**Figure 5.16** This  $I$ - $V$  graph for a light bulb shows that the resistance has increased.



Why do we plot  $V$  on the x-axis? To produce an  $I$ - $V$  graph, the current through the resistor is measured as the p.d. is changed. The variable that is changed is called the independent variable, the variable that changes is the dependent variable. The independent variable is generally plotted on the x-axis; that's why we plot  $I$  against  $V$ . So if an  $I$ - $V$  graph gets steeper it means that the resistance is getting lower.



## Exercises

- 1 If a p.d. of 9V causes a current of 3 mA to flow through a wire, what is the resistance of the wire?
- 2 A current of 1  $\mu$ A flows through a 300 k $\Omega$  resistor. What is the p.d. across the resistor?
- 3 If the p.d. across a 600  $\Omega$  resistor is 12V, how much current flows?
- 4 Below is a table of the p.d. and current through a device called a thermistor.

| p.d./V | Current/A |
|--------|-----------|
| 1.0    | 0.01      |
| 10     | 0.1       |
| 25     | 1.0       |

Calculate the resistance at different potential differences.

## 5.2 Electric circuits

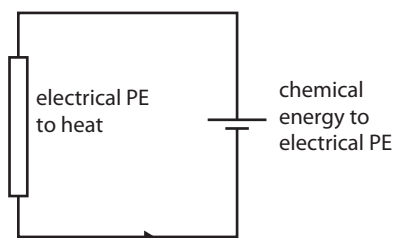
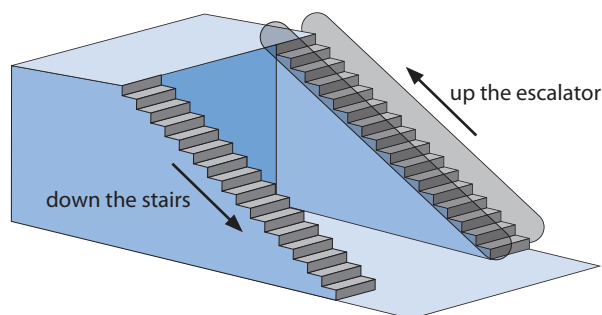
### Assessment statements

- 5.2.1 Define electromotive force (emf).
- 5.2.2 Describe the concept of internal resistance.
- 5.2.3 Apply the equations for resistors in series and in parallel.
- 5.2.4 Draw circuit diagrams.
- 5.2.5 Describe the use of ideal ammeters and ideal voltmeters.
- 5.2.6 Describe a potential divider.
- 5.2.7 Explain the use of sensors in potential divider circuits.
- 5.2.8 Solve problems involving electric circuits.

### The simple circuit – an energy view

We can now consider the energy changes that take place as current flows around the simple circuit.

**Figure 5.17** The shopping centre analogy. The energy changes of the charges that flow round an electric circuit are similar to the energy changes as you walk around a multi-storey shopping centre. The escalator gives you potential energy as it takes you up. When you come down the stairs your potential energy gets less. This energy is given to the stairs; as your feet hit each step, the step gets hot.







The cell uses chemical energy to place the charges in a position of high potential. As the charges flow through the resistor, this potential energy is converted to heat, so the resistor gets hot. These changes in energy are defined by the following terms:

- Emf ( $\mathcal{E}$ )

The emf of a cell is the amount of chemical energy converted to electrical energy, per unit charge. The unit is the volt (V).

- Potential difference (p.d. or V)

The p.d. across a resistance is the amount of electrical energy converted to heat, per unit charge. The unit is the volt (V).

Both of these quantities are energy per unit charge, but emf specifically applies to cells, batteries, generators and any other device that gives the charges potential energy. (One volt is the same as one joule per coulomb.)

### Applying the law of conservation of energy

Since energy cannot be created or destroyed, the energy converted from chemical to electrical in the cell must be equal to the amount converted from electrical to heat in the resistor.

$$\text{From Ohm's law, } V = IR$$

### Worked examples

- 1 If the emf of a battery is 9 V, how much energy is converted from chemical to electrical when 2 C of charge flow?
- 2 What is the p.d. across a resistor if 24 J of heat are produced when a current of 2 A flows through it for 10 s?

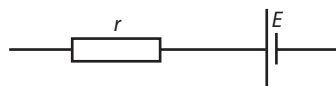
#### Solutions

- 1 emf = energy converted from chemical to electrical per unit charge.  
So energy converted =  $2 \times 9 \text{ J}$   
 $= 18 \text{ J}$
- 2  $2 \text{ A for } 10 \text{ s} = 2 \times 10 \text{ C} = 20 \text{ C of charge.}$   
If 20 C of charge flows, then the energy per unit charge =  $\frac{24}{20} \text{ V} = 1.2 \text{ V}$

## Internal resistance of cells

All cells are made of materials that have resistance. The resistance of the cell is called the *internal resistance*.

If a cell with internal resistance is connected to a resistor, current will flow from the cell. As current flows through the internal resistance, some energy is converted from electrical to heat inside the cell (so the cell gets hot). This means that there is less energy to be converted to heat in the resistor. The p.d. across the resistor is therefore less than the emf of the cell.



To look at the simulations on the PhET website, visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on Weblink 5.4.

'Battery voltage' shows how the battery changes chemical energy to electrical potential energy by moving the charges.

'Battery-resistor current' shows how the electrical energy is converted to heat in the resistor.

Note: these simulations show the movement of electrons, not conventional current.

● **Examiner's hint:** If the amount of charge = 2 C then energy converted =  $2 \times \text{emf}$

● **Examiner's hint:** Current is charge flowing per unit time.

● **Examiner's hint:** From the definition, p.d. = energy converted from electrical to heat per unit charge.

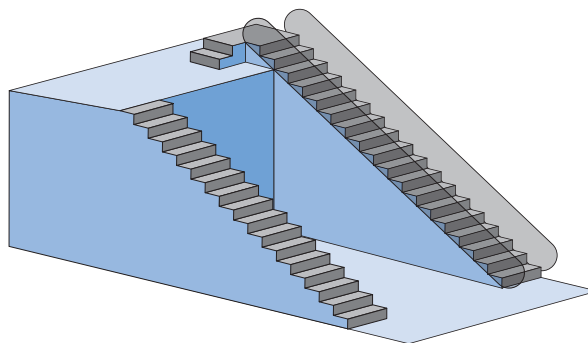


The shopping centre analogy is an example of how we use models to visualize things that we can't see. It is much easier to understand the changes in height as you walk around the shops than the changes of energy as electrons flow around a circuit.

**Figure 5.18** The symbol for a cell includes a resistor  $r$  next to the cell to show internal resistance.



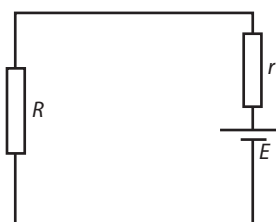
**Figure 5.19** Internal resistance acts like a small step down.



### Shopping centre analogy

In the shopping centre analogy you can see that the elevator still lifts you up to the same level but you must come down a few steps as soon as you step off it.

In the electrical circuit the emf is still the same but the charges lose some energy before they leave the battery.



Applying Ohm's law to the internal resistance, the p.d. across it will be  $Ir$ .

From the law of conservation of energy, when a certain charge flows, the amount of energy converted from chemical to electrical equals the amount converted from electrical to heat.

$$\mathcal{E} = IR + Ir$$

Rearranging this formula, we can get an equation for the current from the battery.

$$I = \frac{\mathcal{E}}{R + r}$$

### Worked example

A battery of emf 9 V with an internal resistance  $1\ \Omega$  is connected to a  $2\ \Omega$  resistor, as shown in Figure 5.20.

How much current will flow?

#### Solution

$$I = \frac{\mathcal{E}}{R + r}$$

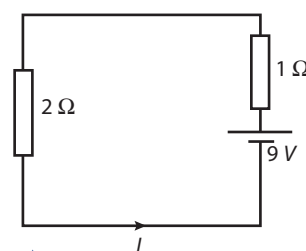
$$\begin{aligned} I &= \frac{9}{2 + 1}\text{ A} \\ &= 3\text{ A} \end{aligned}$$

What is the p.d. across the  $2\ \Omega$  resistor?

$$V = IR$$

$$\begin{aligned} V &= 3 \times 2\text{ V} \\ &= 6\text{ V} \end{aligned}$$

● **Examiner's hint:** The current through the resistor is 3 A.



**Figure 5.20** Always start by drawing a circuit showing the quantities you know and labelling the ones you want to find.

### Exercises

- 5 A current of 0.5 A flows when a battery of emf 6 V is connected to an  $11\ \Omega$  resistor. What is the internal resistance of the battery?
- 6 A 12 V battery with internal resistance  $1\ \Omega$  is connected to a  $23\ \Omega$  resistor. What is the p.d. across the  $23\ \Omega$  resistor?



## Electrical power

Electrical power is the rate at which energy is changed from one form to another.

### Power delivered

In a perfect battery, the power is the amount of chemical energy converted to electrical energy per unit time.

If the emf of a battery is  $\mathcal{E}$ , then if a charge  $Q$  flows, the amount of energy converted from chemical to electrical is  $\mathcal{E}Q$ .

If this charge flows in a time  $t$  then the power delivered  $= \frac{\mathcal{E}Q}{t}$

But  $\frac{Q}{t} =$  the current,  $I$

So the power delivered  $= \mathcal{E}I$

In a real battery, the actual power delivered will be a bit less, since there will be some power dissipated in the internal resistance.

### Power dissipated

The power dissipated in the resistor is the amount of electrical energy converted to heat per unit time.

Consider a resistance  $R$  with a p.d.  $V$  across it. If a charge  $q$  flows in time  $t$  then the current,  $I = \frac{q}{t}$

The p.d.,  $V$ , is defined as the energy converted to heat per unit charge, so the energy converted to heat in this case  $= Vq$

Power is the energy used per unit time, so  $P = \frac{Vq}{t}$

but  $\frac{q}{t} = I$ , so  $P = VI$

### Worked examples

- 1 If a current of 2 A flows through a resistor that has a p.d. of 4 V across it, how much power is dissipated?
- 2 What power will be dissipated when a current of 4 A flows through a resistance of  $55\ \Omega$ ?

### Solutions

- 1  $P = VI$  where  $V = 4\text{ V}$  and  $I = 2\text{ A}$

$$\begin{aligned}P &= 4 \times 2\text{ W} \\&= 8\text{ W}\end{aligned}$$

- 2 Using Ohm's law  $V = IR$

$$\begin{aligned}&= 4 \times 55\text{ V} \\&= 220\text{ V} \\P &= VI \\&= 220 \times 4\text{ W} \\&= 880\text{ W}\end{aligned}$$

First we need to find out the p.d. across the resistor



#### Kilowatt-hours

The electrical energy used at home is measured in kilowatt-hours. This is the amount of energy used by a 1 kilowatt heater switched on for 1 hour. When the electricity bill comes, you have to pay for each kilowatt-hour of energy that you have used.

Since  $1\text{ W} = 1\text{ J s}^{-1}$

1 kilowatt-hour  $= 1000 \times 60 \times 60\text{ J}$

1 kWh  $= 3.6 \times 10^6\text{ J}$



**Power**

$$P = VI$$

$$P = I^2R$$

$$P = V^2/R$$

**Alternative ways of writing  $P = VI$** 

In Example 2, we had to calculate the p.d. before finding the power. It would be convenient if we could solve this problem in one step. We can write alternative forms of the equation by substituting for  $I$  and  $V$  from Ohm's law.

We have shown that power  $P = VI$

But from Ohm's law  $V = IR$

If we substitute for  $V$  we get  $P = IR \times I = I^2R$

We can also substitute for  $I = \frac{V}{R}$

$$\text{Power} = V \times I = \frac{V \times V}{R} = \frac{V^2}{R}$$

**Exercises**

- 7 5 A flows through a  $20\ \Omega$  resistor.
  - (a) How much electrical energy is converted to heat per second?
  - (b) If the current flows for one minute, how much energy is released?
- 8 If a battery has an internal resistance of  $0.5\ \Omega$ , how much power will be dissipated in the battery when 0.25 A flows?
- 9 A current of 0.5 A flows from a battery of emf 9V. If the power delivered is 4 W, how much power is dissipated in the internal resistance?

**Electric kettle (water boiler)**

An electric kettle transfers the heat produced when current flows through a wire element to the water inside the kettle.

**Worked example**

A current of 3 A flows through an electric kettle connected to the 220 V mains. What is the power of the kettle and how long will it take to boil 1 litre of water?

**Solution**

The power of the kettle =  $VI = 220 \times 3 = 660\ \text{W}$

To calculate energy needed to boil the water, we use the formula

$$\text{heat required} = \text{mass} \times \text{specific heat capacity} \times \text{temperature change.}$$

The specific heat capacity of water is  $4180\ \text{J kg}^{-1}\ ^\circ\text{C}^{-1}$

The mass of 1 litre of water is 1 kg, so if we assume that the water was at room temperature,  $20^\circ\text{C}$ , then to raise it to  $100^\circ\text{C}$  the energy required is:

$$1 \times 4180 \times 80 = 334\ 400\ \text{J}$$

$$\begin{aligned} \text{power} &= \text{energy/time, so the time taken} = \frac{\text{energy}}{\text{power}} \\ &= \frac{334\ 400}{660} \end{aligned}$$

$$\text{Time} = 506.67\ \text{s}$$

$$= 8\ \text{minutes}\ 27\ \text{seconds}$$

**The light bulb**

If the power dissipated in a wire is large, then a lot of heat is produced per second. When heat is added quickly, the wire doesn't have time to lose this heat to the surroundings. The result is that the temperature of the wire increases, and if the

temperature is high enough, the wire will begin to glow, giving out light. Only about 10% of the energy dissipated in a light bulb is converted to light - the rest is heat.

## The electric motor

A motor converts electrical energy to mechanical energy; this could be in the form of potential energy, if something is lifted by the motor, or kinetic energy, if the motor is accelerating something like a car.

## Worked example

An electric motor is used to lift 10 kg through 3 m in 5 seconds. If the p.d. across the motor is 12 V, how much current flows (assuming no energy is lost)?

### Solution

$$\begin{aligned}\text{Work done by the motor} &= mgh \\ &= 10 \times 10 \times 3 \text{ J} \\ &= 300 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{Power} &= \frac{\text{work done}}{\text{time}} \\ &= \frac{300}{5} \text{ W} \\ &= 60 \text{ W}\end{aligned}$$

$$\begin{aligned}\text{Electrical power, } P &= IV \text{ so } I = P/V \\ &= \frac{60}{12} \text{ A} \\ &= 5 \text{ A}\end{aligned}$$



Fluorescent tubes are much more efficient than incandescent light bulbs, converting most of the electrical energy to light.

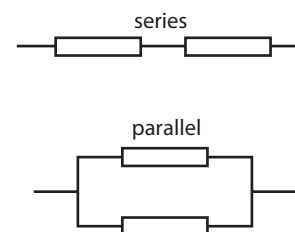
## Exercises

- 10** An electric car of mass 1000 kg uses twenty-five 12 V batteries connected together to create a p.d. of 300 V. The car accelerates from rest to a speed of  $30 \text{ m s}^{-1}$  in 12 seconds.
- What is the final kinetic energy of the car?
  - What is the power of the car?
  - How much electrical current flows from the battery?
- What assumptions have you made in calculating **a)** to **c)**?
- 11** A light bulb for use with the 220 V mains is rated at 100 W.
- What current will flow through the bulb?
  - If the bulb converts 20% of the energy to light, how much light energy is produced per second?
- 12** A 1 kW electric heater is connected to the 220 V mains and left on for 5 hours.
- How much current will flow through the heater?
  - How much energy will the heater release?

## Combinations of components

In practical situations, resistors and cells are often joined together in combinations e.g. fairy lights, flashlight batteries.

There are many ways of connecting a number of components – we will consider two simple arrangements, series and parallel.

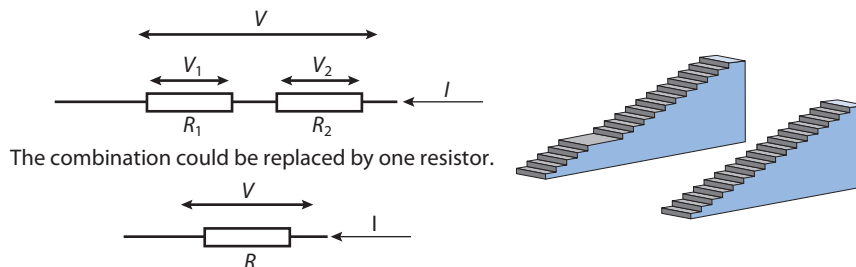


**Figure 5.21** Two simple combinations of resistors.

## Resistors in series

In a series circuit the same current flows through each resistor.

**Figure 5.22** Two resistors in series are similar to two flights of stairs.



Applying the law of conservation of energy, the p.d. across  $R_1$  plus the p.d. across  $R_2$  must be equal to the p.d. across the combination.

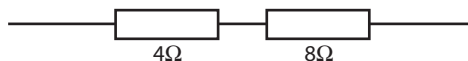
$$V_1 + V_2 = V$$

Applying Ohm's law to each resistor:  $IR_1 + IR_2 = IR$

Dividing by  $I$ :  $R_1 + R_2 = R$

## Worked example

What is the total resistance of a  $4\Omega$  and an  $8\Omega$  resistor in series?



**Figure 5.23**

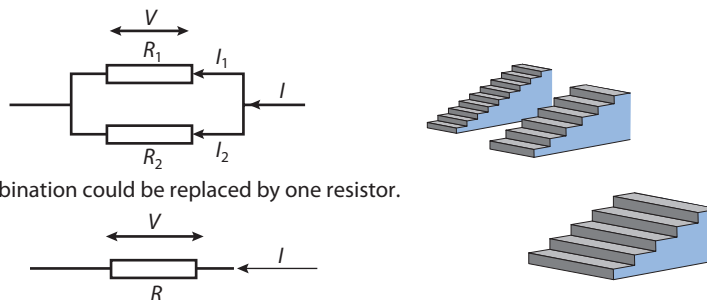
### Solution

$$\begin{aligned}\text{Total resistance} &= R_1 + R_2 \\ &= 4 + 8 \Omega \\ &= 12 \Omega\end{aligned}$$

## Resistors in parallel

In a parallel circuit the current splits in two.

**Figure 5.24** Resistors in parallel are similar to stairs side by side.



Applying the law of conservation of charge, we know that the current going into a junction must equal the current coming out.

$$I = I_1 + I_2$$

Applying Ohm's law to each resistor gives:  $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2}$

Dividing by  $V$ :  $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

## Worked examples

- 1 What is the total resistance of a  $4\ \Omega$  and an  $8\ \Omega$  resistor in parallel?
- 2 What is the total resistance of two  $8\ \Omega$  resistors in parallel?

### Solutions

1 Using  $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{8}$$

$$= \frac{2+1}{8} = \frac{3}{8}$$

$$\text{so } R = \frac{8}{3}\ \Omega$$

2 Using  $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

$$\frac{1}{R} = \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$$

$$R = \frac{8}{2}\ \Omega$$

$$= 4\ \Omega$$

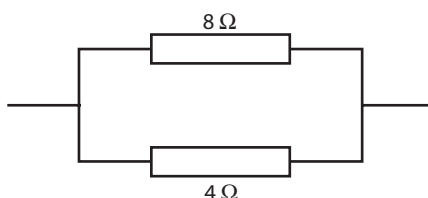


Figure 5.25

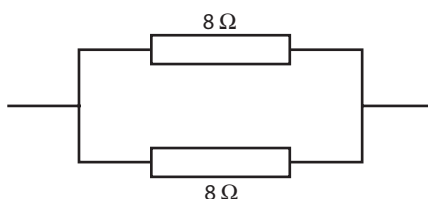


Figure 5.26 The total resistance of two equal resistors in parallel is half the resistance of one of them.

## Multiple combinations

When there are many resistors, the total is found by splitting the circuit into small units of parallel and series resistors; for example in the circuit shown in Figure 5.27.

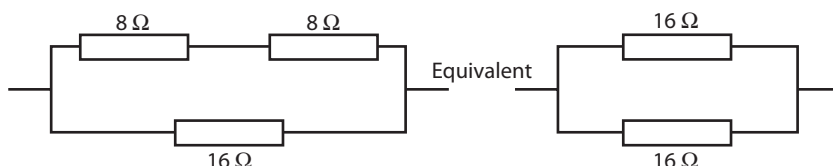


Figure 5.27 Equivalent circuits.

The two  $8\ \Omega$  resistors at the top are equivalent to one  $16\ \Omega$  resistor.

This gives two  $16\ \Omega$  resistors in parallel, a total of  $8\ \Omega$ .

## Exercises

Calculate the total resistance for the circuits in Figure 5.28.

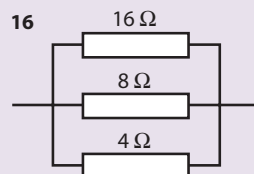
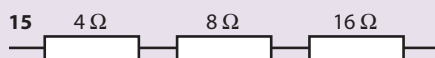
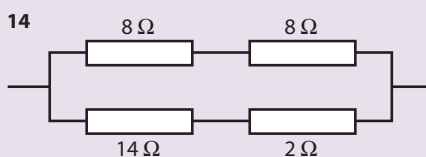
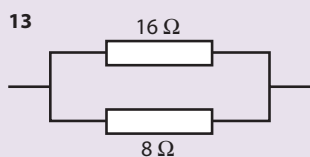


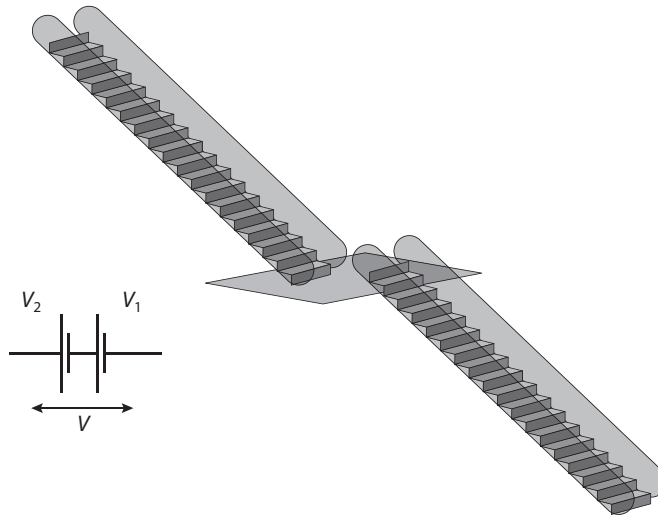
Figure 5.28



### Cells in series

Cells are very often added in series to produce a larger p.d.

**Figure 5.29** Cells in series are similar to two flights of escalators.



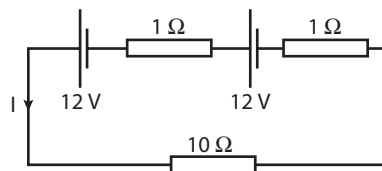
The p.d. of cells in series simply add

$$V = V_1 + V_2$$

The three resistors are in series (it looks like the  $10\ \Omega$  is in parallel with the  $1\ \Omega$  resistors, but they are in series because the same current flows through them all).



**Figure 5.30** The circuit diagram.



The total p.d. for two batteries in series =  $12 + 12 = 24\text{ V}$

### Solution

Total resistance =  $1 + 1 + 10 = 12\ \Omega$ .

Applying Ohm's law,

$$I = \frac{V}{R}$$

$$= \frac{24}{12}\text{ A}$$

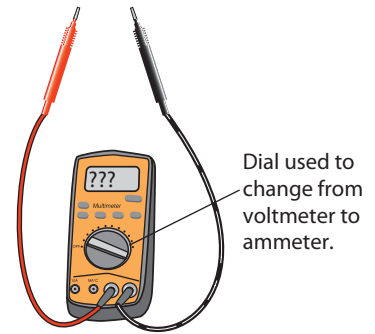
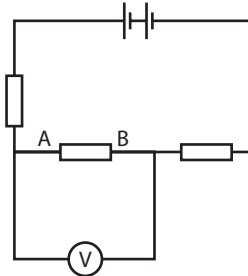
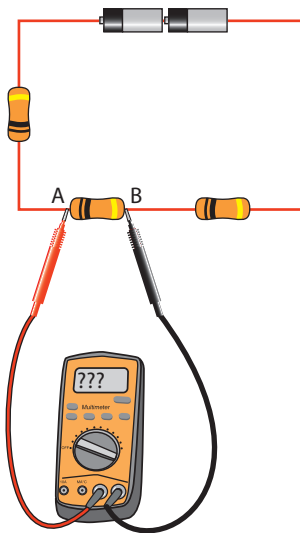
$$= 2\text{ A}$$

## Electrical measurement

### Measurement of potential difference

The p.d. can be measured using a voltmeter. There are two main types of voltmeter, digital and analogue.

The p.d. is the difference in potential between two points. To measure the p.d. between A and B, one lead of the voltmeter must be connected to A, the other to B.



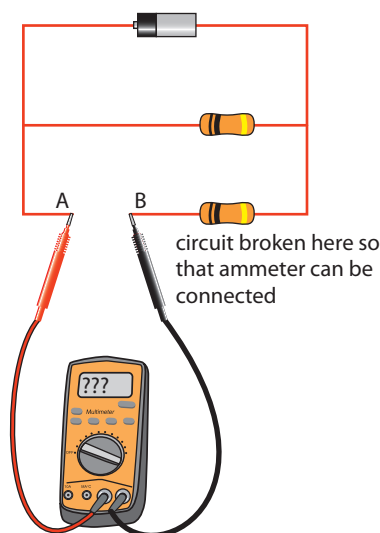
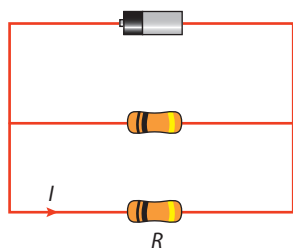
**Figure 5.31** A multimeter is a common instrument that can measure both p.d. and current. It can also measure resistance.

**Figure 5.32** A voltmeter is connected from A to B.

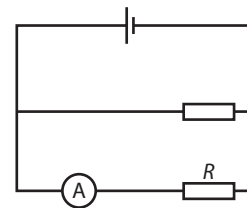
An ideal voltmeter has infinitely high resistance so that it does not take any current from the circuit.

### Measurement of current

To measure the current flowing through a resistor, the ammeter must be connected so that the same current will flow through the ammeter as flows through the resistor. This means disconnecting one of the wires and connecting the ammeter.



circuit broken here so that ammeter can be connected

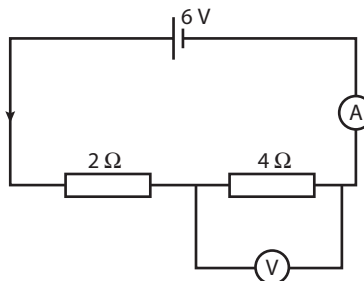


**Figure 5.33** The ammeter is connected to measure the current through  $R$ .

An ideal ammeter has zero resistance so that it doesn't change the current in the circuit.

**Worked example**

Calculate the current and potential difference measured by the meters in the circuit in Figure 5.34. Assume the battery has no internal resistance and that the meters are ideal.

**Solution**

$$\text{Total resistance} = 2 + 4 \Omega$$

$$= 6 \Omega$$

The two resistors are in series.

$$I = \frac{V}{R}$$

Applying Ohm's law, to find the current.

$$= \frac{6}{6} \text{ A}$$

$$= 1 \text{ A}$$

So ammeter reading = 1 A

$$V = IR$$

Applying Ohm's law to the 4 Ω resistor.

$$= 1 \times 4 \text{ V}$$

$$= 4 \text{ V}$$

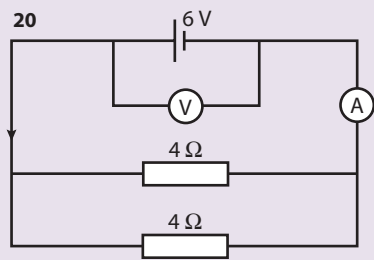
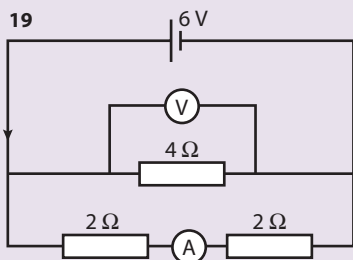
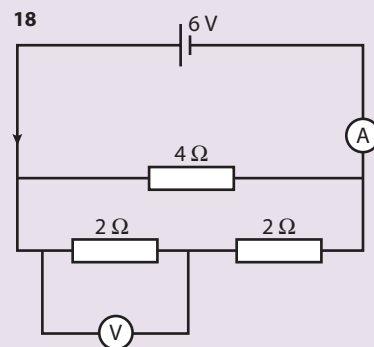
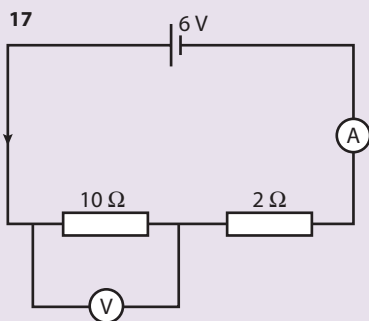
So the voltmeter reads 4 V.

Figure 5.34

**Exercises**

Find the ammeter and voltmeter readings in the circuits in Figure 5.35. All meters are ideal and the batteries have no internal resistance. (You can build them in the PhET 'circuit construction kit' to see whether your answers agree.)

Figure 5.35



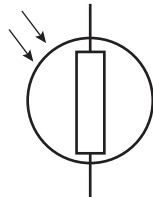
## Electrical sensors

An electrical sensor is a device whose electrical properties change with changing physical conditions.

### Thermistor

A thermistor is made from a semiconducting material whose resistance decreases as temperature increases. As the thermistor gets hotter, more charge carriers are released, so the current can flow more easily.

### Light sensor (LDR)



A light sensor or light-dependent resistor (LDR) is also a semiconducting device, but, unlike the thermistor, it is light that releases more charge carriers, resulting in a lower resistance.

### Strain gauge

A strain gauge is a thin metal wire. If it is stretched, its length increases and its cross-sectional area gets smaller. This results in an increase in resistance.

## Use of sensors

The resistance of all three sensors varies with some physical property. However, it would be much more useful if the devices gave a changing p.d. rather than a changing resistance. To convert the changing resistance to a changing p.d., we use a potential divider.

### The potential divider

The battery creates a p.d. across the resistors equal to  $V_{in}$

From Ohm's law we know that the current in the circuit  $I = \frac{V}{R}$

Since the total resistance is  $R_1 + R_2$

The current  $I = \frac{V_{in}}{R_1 + R_2}$  Equation (1)

The p.d. across  $R_2 = V_{out}$

Applying Ohm's law to  $R_2$  gives

$$V_{out} = IR_2$$

Substituting from equation (1) gives

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

This is the potential divider formula.

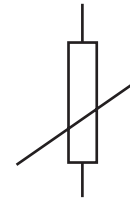
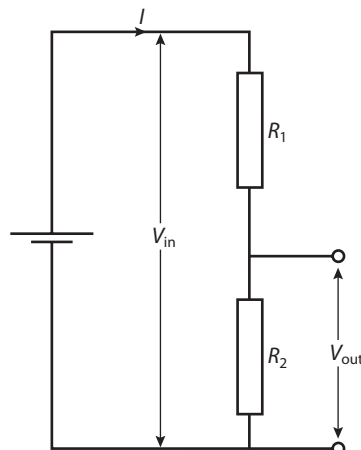


Figure 5.36 A thermistor.

Figure 5.37 An LDR.

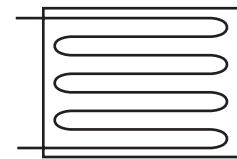
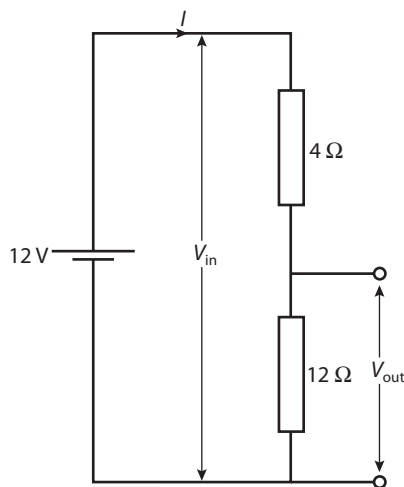


Figure 5.38 A strain gauge.

Figure 5.39 A potential divider circuit consists of two series resistors.

**Worked example**

Calculate the output voltage for the potential divider in Figure 5.40.

**Figure 5.40****Solution**

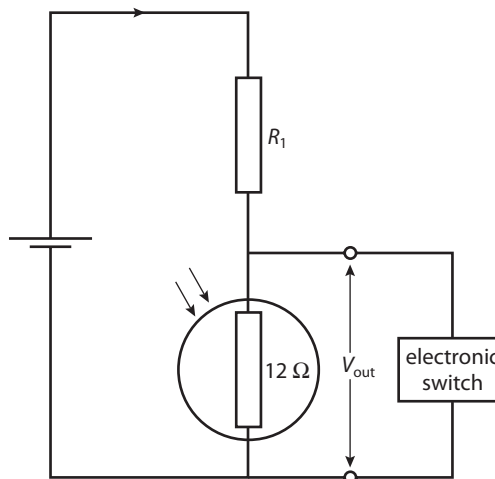
Using the potential divider formula

$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$$

$$V_{\text{out}} = 12 \frac{12}{4 + 12} = 9 \text{ V}$$

**Using the potential divider with sensors****Automatic light switch**

**Figure 5.41** An LDR and a potential divider can be used to operate an automatic light switch.



When light stops shining on the LDR, its resistance increases, resulting in an increase in  $V_{\text{out}}$ . The increase in  $V_{\text{out}}$  in turn activates the electronic switch that puts on the lights. The electronic switch needs a minimum p.d. to activate it, so it doesn't switch on the lights until  $V_{\text{out}}$  is big enough.

The important point here is that the electronic switch needs a p.d. to activate it, hence the need for a potential divider.

## Worked example

The battery in Figure 5.41 has an emf of 12 V and no internal resistance. The p.d. required to activate the switch is 5 V. Find the value of  $R_1$  that will cause the lights to turn on when the resistance of the LDR rises to 200 k $\Omega$ .

### Solution

$$V_{\text{in}} = 12 \text{ V}$$

$$R_2 = 200 \text{ k}\Omega$$

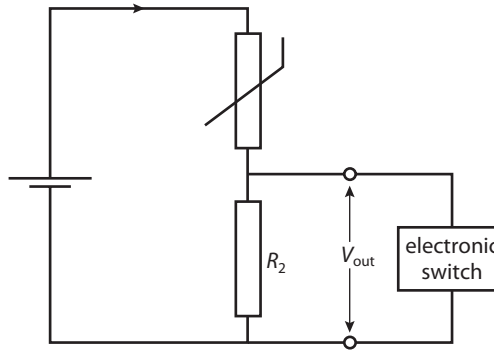
$$V_{\text{out}} = 5 \text{ V}$$

Rearranging the potential divider equation  $V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$

$$\begin{aligned} \text{gives } R_1 &= R_2 \frac{V_{\text{in}} - V_{\text{out}}}{V_{\text{out}}} \\ &= 200 \text{ k}\Omega \times \frac{(12 - 5)}{5} \\ &= 280 \text{ k}\Omega \end{aligned}$$

## The fire alarm

When the thermistor gets hot, its resistance decreases, resulting in an increased current through  $R_2$ , which in turn leads to an increase in  $V_{\text{out}}$ . The increase in  $V_{\text{out}}$  activates an electronic switch that rings a bell.



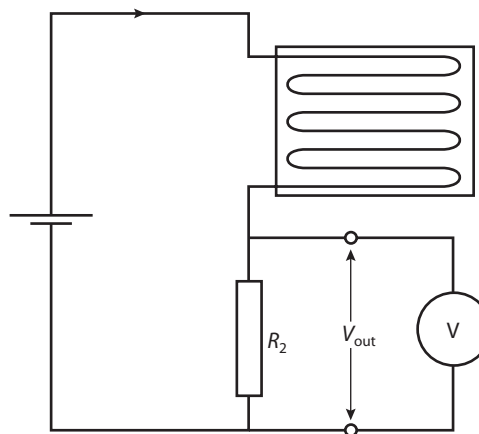
**Figure 5.42** The fire alarm bell must ring if the thermistor gets hot.

## Exercises

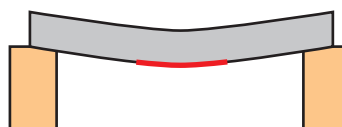
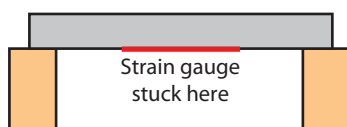
- 21** Assume that the circuit in Figure 5.42 has a 12 V battery and a switch that activates when the p.d. is 5 V. Calculate the value of  $R_2$ , if the bell rings when the resistance of the thermistor drops to 1 k $\Omega$ ?

## Using a strain gauge

A strain gauge can be used to detect whether parts of a building are stretching. For example, a strain gauge stuck to the underside of a bridge will be stretched if the bridge bends when a heavy truck crosses it. If the strain gauge is connected in a potential divider circuit, the  $V_{\text{out}}$  can be used to measure how much the bridge stretches.



**Figure 5.43** Strain gauge circuit.

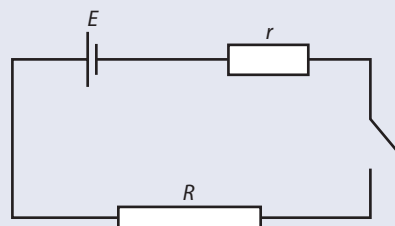


**Figure 5.44** As the bridge bends, the strain gauge gets longer.

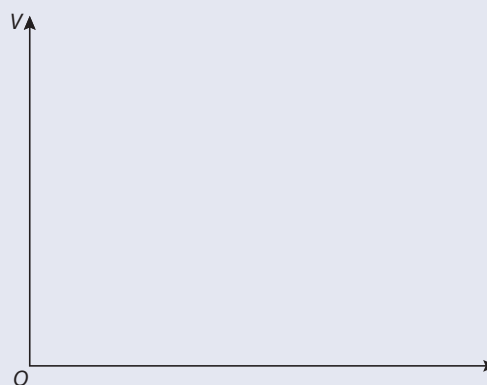
## Practice questions

1 This question is about electrical energy and associated phenomena.

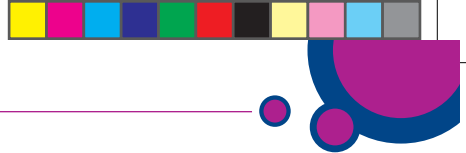
A cell of emf  $E$  and internal resistance  $r$  is connected in series with a resistor  $R$ , as shown below. The cell supplies  $8.1 \times 10^3 \text{ J}$  of energy when  $5.8 \times 10^3 \text{ C}$  of charge moves completely round the circuit. The current in the circuit is constant.



- (i) Calculate the emf  $E$  of the cell. (2)
  - (ii) The resistor  $R$  has resistance  $6.0 \Omega$ . The potential difference between its terminals is  $1.2 \text{ V}$ . Determine the internal resistance  $r$  of the cell. (3)
  - (iii) Calculate the total energy transfer in the resistor  $R$ . (2)
  - (iv) Describe, in terms of a simple model of electrical conduction, the mechanism by which the energy transfer in the resistor  $R$  takes place. (5)
- 2 This question is about a filament lamp.
- (a) On the axes below, draw a sketch graph to show the variation with potential difference  $V$  of the current  $I$  in a typical filament lamp (the  $I$ - $V$  characteristic). (Note: this is a sketch graph; you do not need to add any values to the axes). (1)

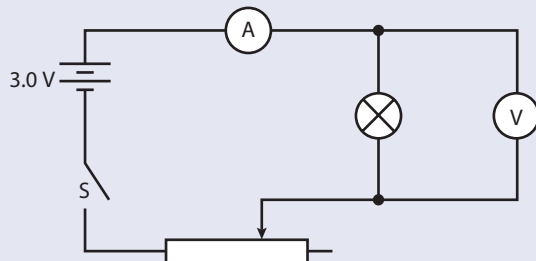


- (b) (i) Explain how the resistance of the filament is determined from the graph. (1)
  - (ii) Explain whether the graph you have sketched indicates ohmic behaviour or non-ohmic behaviour. (1)
- A filament lamp operates at maximum brightness when connected to a  $6.0 \text{ V}$  supply. At maximum brightness, the current in the filament is  $120 \text{ mA}$ .
- (c) (i) Calculate the resistance of the filament when it is operating at maximum brightness. (1)
  - (ii) You have available a  $24 \text{ V}$  supply and a collection of resistors of a suitable power rating and with different values of resistance. Calculate the resistance of the resistor that is required to be connected in series with the supply such that the voltage across the filament lamp will be  $6.0 \text{ V}$ . (2)



3 This question is about electric circuits.

Susan sets up the circuit below in order to measure the current–voltage ( $I$ – $V$ ) characteristic of a small filament lamp.



The supply is a battery that has an emf of 3.0 V and the ammeter and voltmeter are considered to be ideal. The lamp is labelled by the manufacturer as ‘3 volts, 0.6 watts’.

- (a) (i) Explain what information this labelling provides about the normal operation of the lamp. (2)
- (ii) Calculate the current in the filament of the lamp when it is operating at normal brightness. (2)

Susan sets the variable resistor to its maximum value of resistance. She then closes the switch S and records the following readings.

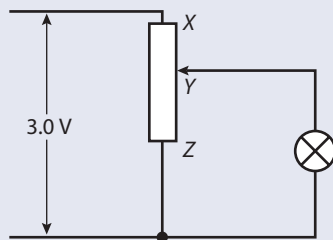
Ammeter reading = 0.18 A      Voltmeter reading = 0.60 V

She then sets the variable resistor to its zero value of resistance and records the following readings.

Ammeter reading = 0.20 A      Voltmeter reading = 2.6 V

- (b) (i) Explain why, by changing the value of the resistance of the variable resistance, the potential difference across the lamp cannot be reduced to zero or be increased to 3.0 V. (2)
- (ii) Determine the internal resistance of the battery. (3)
- (c) Calculate the resistance of the filament when the reading on the voltmeter is
- (i) 0.60 V. (1)
- (ii) 2.6 V. (1)
- (d) Explain why there is a difference between your answers to (c) (i) and (c) (ii). (2)
- (e) Using the axes as in question 2, draw a sketch-graph of the  $I$ – $V$  characteristic of the filament of the lamp. (Note: this is a sketch-graph; you do not need to add any values to the axis.) (1)

The diagram below shows an alternative circuit for varying the potential difference across the lamp.



The potential divider XZ has a potential of 3.0 V across it. When the contact is at the position Y, the resistance of XY equals the resistance of YZ which equals  $12\ \Omega$ . The resistance of the lamp is  $4\ \Omega$ .

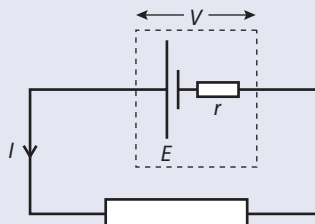
- (f) Calculate the potential difference across the lamp. (4)



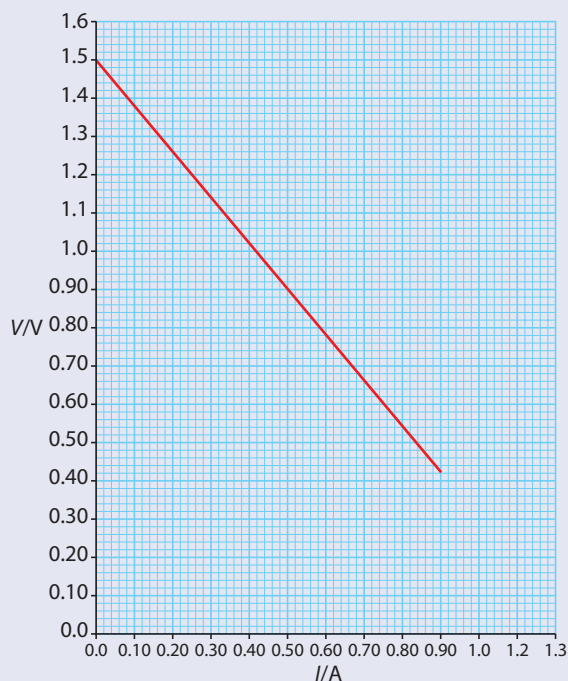


4 This question is about emf and internal resistance.

A dry cell has an emf  $E$  and internal resistance  $r$  and is connected to an external circuit. There is a current  $I$  in the circuit when the potential difference across the terminals of the cell is  $V$ .



- (a) State expressions, in terms of  $E$ ,  $V$ ,  $r$  and  $I$  where appropriate, for
- (i) the total power supplied by the cell. (1)
  - (ii) the power dissipated in the cell. (1)
  - (iii) the power dissipated in the external circuit. (1)
- (b) Use your answers to (a) to derive a relationship between  $V$ ,  $E$ ,  $I$  and  $r$ . (2)
- The graph below shows the variation of  $V$  with  $I$  for the dry cell.



- (c) Draw the circuit that could be used to obtain the data from which the graph was plotted. (3)
- (d) Use the graph, explaining your answers, to
- (i) determine the emf  $E$  of the cell. (2)
  - (ii) determine the current in the external circuit when the resistance  $R$  of the external circuit is very small. (2)
  - (iii) deduce that the internal resistance  $r$  of the cell is about  $1.2\ \Omega$ . (1)
- (e) The maximum power dissipated in the external circuit occurs when the resistance of the external circuit has the same value as the internal resistance of the cell. Calculate the maximum power dissipation in the external circuit. (3)

## 6.5 Electric force and field

### Assessment statements

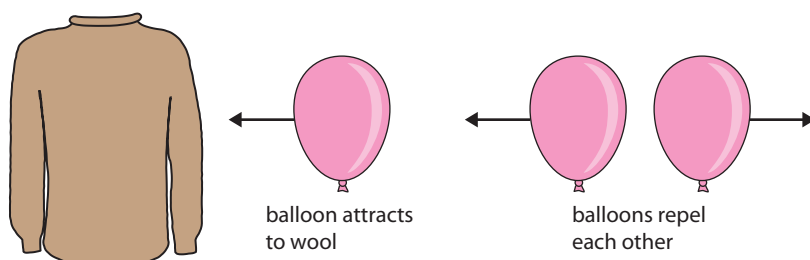
- 6.2.1 State that there are two types of electric charge.
- 6.2.2 State and apply the law of conservation of charge.
- 6.2.3 Describe and explain the difference in the electrical properties of conductors and insulators.
- 6.2.4 State Coulomb's law.
- 6.2.5 Define *electric field strength*.
- 6.2.6 Determine the electric field strength due to one or more point charges.
- 6.2.7 Draw the electric field patterns for different charge configurations.
- 6.2.8 Solve problems involving electric charges, forces and fields.

### Electric force

So far we have dealt with many forces; for example, friction, tension, upthrust, normal force, air resistance and gravitational force. If we rub a balloon on a woollen pullover, we find that the balloon is attracted to the wool of the pullover – this cannot be explained in terms of any of the forces we have already considered, so we need to develop a new model to explain what is happening. First we need to investigate the effect.

Consider a balloon and a woollen pullover – if the balloon is rubbed on the pullover, we find that it is attracted to the pullover. However, if we rub two balloons on the pullover, the balloons repel each other.

Whatever is causing this effect must have two different types, since there are two different forces. We call this force the *electric force*.



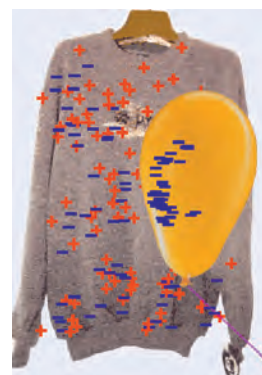
### Charge

The balloon and pullover must have some property that is causing this force. We call this property *charge*. There must be two types of charge, traditionally called positive (+ve) and negative (–ve). To explain what happens, we can say that, when rubbed, the balloon gains –ve charge and the pullover gains +ve charge. If like charges repel and unlike charges attract, then we can explain why the balloons repel and the balloon and pullover attract.



This is another example of how models are used in physics.

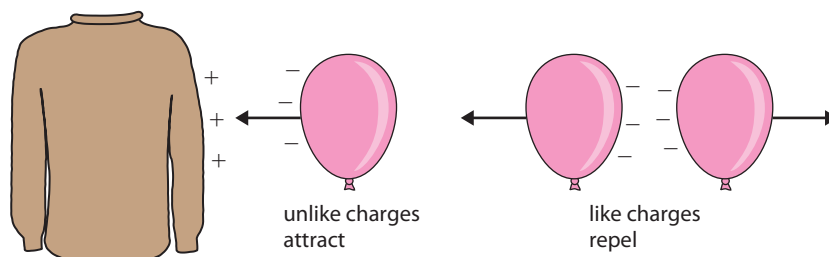
**Figure 6.22** Balloons are attracted to the wool but repel each other.



You can try this with real balloons or, to use the simulation 'Balloons and static electricity', visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on Weblink 6.3.

**Figure 6.23** The force is due to charges.

Here +ve and -ve numbers are used to represent something that they were not designed to represent.

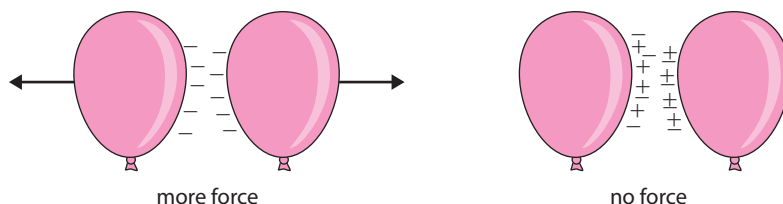


The unit of charge is the coulomb (C).

### Conservation of charge

If we experiment further, we find that if we rub the balloon more, then the force between the balloons is greater. We also find that if we add +ve charge to an equal -ve charge, the charges cancel.

**Figure 6.24** Charges cancel each other out.



We can add and take away charge but we cannot destroy it.

The law of conservation of charge states that charge can neither be created nor destroyed.

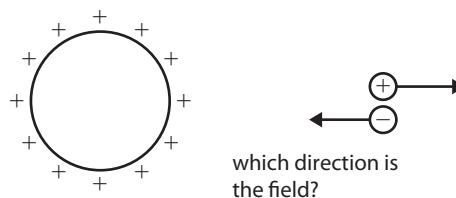
### Electric field

We can see that there are certain similarities with the electric force and gravitational force; they both act without the bodies touching each other. We used the concept of a field to model gravitation and we can use the same idea here.

Electric field is defined as a region of space where a charged object experiences a force due to its charge.

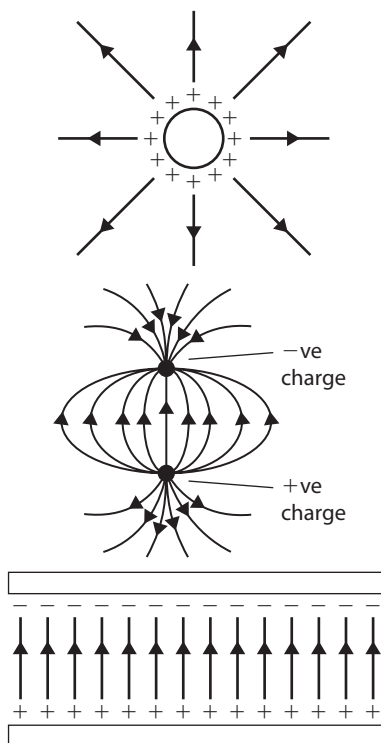
### Field lines

**Figure 6.25** Field is in the direction in which a +ve charge would accelerate.



Field lines can be used to show the direction and strength of the field. However, because there are two types of charge, the direction of the force could be one of two possibilities.

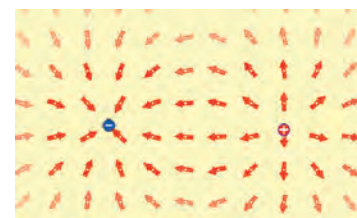
It has been decided that we should take the direction of the field to be the direction that a small +ve charge would accelerate if placed in the field. So we will always consider what would happen if +ve charges are moved around in the field. The field lines will therefore be as shown in Figures 6.26 to 6.28.



**Figure 6.26** Field lines close to a sphere of charge.

**Figure 6.27** Field due to a dipole.

**Figure 6.28** A uniform field.



**W** In the PhET simulation 'Charges and fields' you can investigate the force experienced by a small charge as it is moved around an electric field. To try this, visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on Weblink 6.4.

## Coulomb's law

In a gravitational field, the force between masses is given by Newton's law, and the equivalent for an electric field is Coulomb's law.

Coulomb's law states that the force experienced by two point charges is directly proportional to the product of their charge and inversely proportional to the square of their separation.

The force experienced by two point charges  $Q_1$  and  $Q_2$  separated by a distance  $r$  in a vacuum is given by the formula

$$F = k \frac{Q_1 Q_2}{r^2}$$

The constant of proportionality  $k = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$

Note: Similarly to gravitational fields, Coulomb's law also applies to spheres of charge, the separation being the distance between the centres of the spheres.

## Electric field strength ( $E$ )

The electric field strength is a measure of the force that a +ve charge will experience if placed at a point in the field. It is defined as the force per unit charge experienced by a small +ve test charge placed in the field.

So if a small +ve charge  $q$  experiences a force  $F$  in the field, then the field strength at that point is given by  $E = \frac{F}{q}$ . The unit of field strength is  $\text{NC}^{-1}$ , and it is a vector quantity.

## Worked examples

A  $5 \mu\text{C}$  point charge is placed 20 cm from a  $10 \mu\text{C}$  point charge.

- 1 Calculate the force experienced by the  $5 \mu\text{C}$  charge.
- 2 What is the force on the  $10 \mu\text{C}$  charge?
- 3 What is the field strength 20 cm from the  $10 \mu\text{C}$  charge?

● **Examiner's hint:** Field strength is defined as the force per unit charge so if the force on a  $5\text{ }\mu\text{C}$  charge is  $11.25\text{ N}$ , the field strength  $E$  is equal to  $11.25\text{ N}$  divided by  $5\text{ }\mu\text{C}$ .

● **Examiner's hint:** When solving field problems you always assume one of the charges is in the field of the other. E.g in Example 1 the  $5\text{ }\mu\text{C}$  charge is in the field of the  $10\text{ }\mu\text{C}$  charge. Don't worry about the fact that the  $5\text{ }\mu\text{C}$  charge also creates a field – that's not the field you are interested in.

### Solutions

1 Using the equation  $F = \frac{kQ_1Q_2}{r^2}$

$$Q_1 = 5 \times 10^{-6}\text{ C}, Q_2 = 10 \times 10^{-6}\text{ C and } r = 0.20\text{ m}$$

$$F = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 10 \times 10^{-6}}{0.20^2}\text{ N}$$

$$= 11.25\text{ N}$$

2 According to Newton's third law, the force on the  $10\text{ }\mu\text{C}$  charge is the same as the  $5\text{ }\mu\text{C}$ .

3 Force per unit charge  $= \frac{11.25}{5 \times 10^{-6}}$

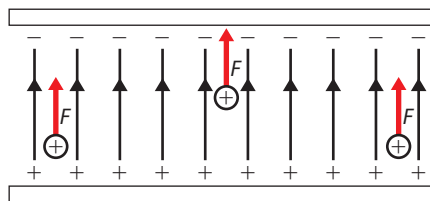
$$E = 2.25 \times 10^6\text{ N C}^{-1}$$

### Exercises

- 20 If the charge on a  $10\text{ cm}$  radius metal sphere is  $2\text{ }\mu\text{C}$ , calculate
- the field strength on the surface of the sphere
  - the field strength  $10\text{ cm}$  from the surface of the sphere
  - the force experienced by a  $0.1\text{ }\mu\text{C}$  charge placed  $10\text{ cm}$  from the surface of the sphere.
- 21 A small sphere of mass  $0.01\text{ kg}$  and charge  $0.2\text{ }\mu\text{C}$  is placed at a point in an electric field where the field strength is  $0.5\text{ N C}^{-1}$ .
- What force will the small sphere experience?
  - If no other forces act, what is the acceleration of the sphere?

### Electric field strength in a uniform field

A uniform field can be created between two parallel plates of equal and opposite charge as shown in Figure 6.29. The field lines are parallel and equally spaced. If a test charge is placed in different positions between the plates, it experiences the same force.



**Figure 6.29** In a uniform field the force on a charge  $q$  is the same everywhere.

So if a test charge  $q$  is placed in the field above, then  $E = \frac{F}{q}$  everywhere between the two charged plates.

### Worked example

If a charge of  $4\text{ }\mu\text{C}$  is placed in a uniform field of field strength  $2\text{ N C}^{-1}$  what force will it experience?

#### Solution

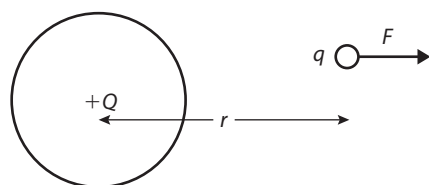
$$F = EQ$$

Rearranging the formula  $E = \frac{F}{Q}$

$$= 2 \times 4 \times 10^{-6}\text{ N}$$

$$= 8\text{ }\mu\text{N}$$

## Electric field strength close to a sphere of charge



From definition:

$$E = \frac{F}{q}$$

From Coulomb's law:

$$F = k \frac{Qq}{r^2}$$

Substituting:

$$E = k \frac{Q}{r^2}$$

## Addition of field strength

Field strength is a vector, so when the field from two negatively charged bodies act at a point, the field strengths must be added vectorially. In Figure 6.31, the resultant field at two points A and B is calculated. At A the fields act in the same line but at B a triangle must be drawn to find the resultant.

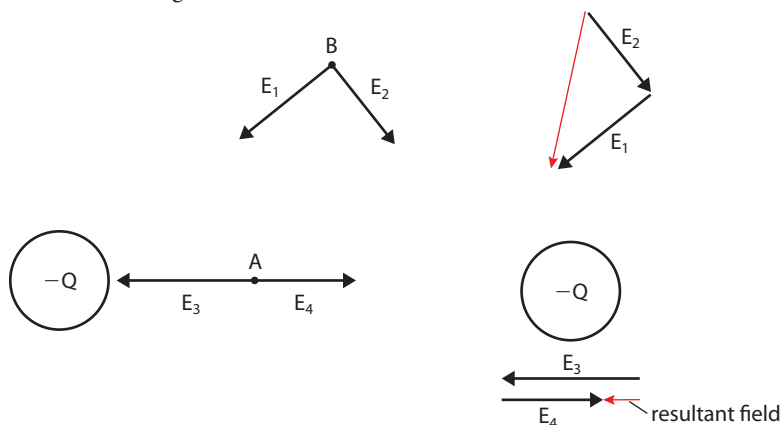
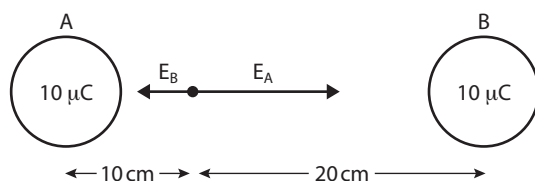


Figure 6.30

Figure 6.31 Since both charges are negative, the field strength is directed towards the charges. Since they are at right angles to each other, Pythagoras can be used to sum these vectors.

$E_3$  is the field due to the charge on the left,  $E_4$  is due to the charge on the right.  $E_3$  is bigger than  $E_4$  because the charge on the left is closer.

## Worked example



Two  $+10 \mu\text{C}$  charges are separated by 30 cm. What is the field strength between the charges 10 cm from A?

### Solution

$$\text{Field strength due to A, } E_A = \frac{9 \times 10^9 \times 10 \times 10^{-6}}{0.1^2}$$

$$= 9 \times 10^6 \text{ N C}^{-1}$$

$$E_B = \frac{9 \times 10^9 \times 10 \times 10^{-6}}{0.2^2}$$

$$= 2.25 \times 10^6 \text{ N C}^{-1}$$

$$\text{Resultant field strength} = (9 - 2.25) \times 10^6 \text{ N C}^{-1}$$

$$= 6.75 \times 10^6 \text{ N C}^{-1} \quad (\text{to the left})$$

## 6.6 Electrical potential

### Assessment statements

- 9.3.1 Define electric potential and electric potential energy.
- 9.3.2 State and apply the expression for electric potential due to a point charge.
- 9.3.3 State and apply the formula relating electric field strength to electric potential gradient.
- 9.3.4 Determine the potential due to one or more point charges.
- 9.3.5 Describe and sketch the pattern of equipotential surfaces due to one and two point charges.
- 9.3.6 State the relation between equipotential surfaces and electric field lines.
- 9.3.7 Solve problems involving electric potential energy and electric potential.

The concept of electric potential is very similar to that of gravitational potential; it gives us information about the amount of energy associated with different points in a field. We have already defined electric potential difference in relation to electrical circuits; it is the amount of electrical energy converted to heat when a unit charge flows through a resistor. In this section we will define the electric potential in more general terms.

### Electric potential energy and potential

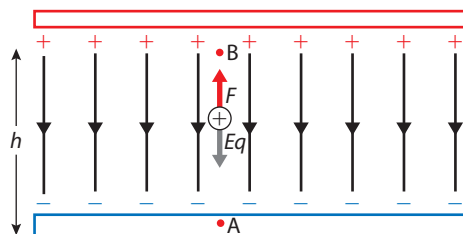


Figure 6.32

When we move a positive charge around in an electric field we have to do work on it. If we do work we must give it energy. This energy is not increasing the KE of the particle so must be increasing its PE, and so we call this *electric potential energy*. Let us first consider the uniform field shown in Figure 6.32. In order to move a charge  $+q$  from A to B, we need to exert a force that is equal and opposite to the electric force,  $Eq$ . As we move the charge we do an amount of work equal to  $Eqh$ . We have therefore increased the electrical PE of the charge by the same amount, so  $PE = Eqh$ .

This is very similar to the room in Figure 6.9; the higher we lift the positive charge, the more PE it gets. In the same way we can define the potential of different points as being the quantity that defines how much energy a given charge would have if placed there.

#### Positive charge

Note that the potential is defined in terms of a positive charge.



**Electric potential at a point is the amount of work per unit charge needed to take a small positive test charge from a place of zero potential to the point.**

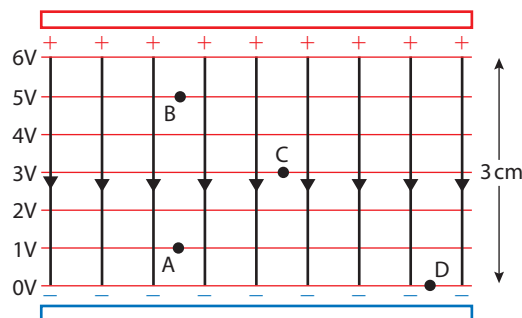
The unit of potential is  $\text{J C}^{-1}$  or volts.

Potential is a scalar quantity.

In this example if we define the zero in potential as the bottom plate, then the potential at B,  $V_B = Eh$

Since the potential is proportional to  $h$  we can deduce that all points a distance  $h$  from the bottom plate will have the same potential. We can therefore draw lines

of equipotential as we did in the gravitational field. Figure 6.33 shows an example with equipotentials.



**Figure 6.33**



### The electronvolt (eV)

This is a unit of energy used in atomic physics. 1 eV = energy gained by an electron accelerated through a p.d. of 1 V.

## Exercises

Refer to Figure 6.33 for Questions 22–27.

- 22** What is the potential difference (p.d.) between A and C?
- 23** What is the p.d. between B and D?
- 24** If a charge of +3 C was placed at B, how much PE would it have?
- 25** If a charge of +2 C was moved from C to B, how much work would be done?
- 26** If a charge of −2 C moved from A to B, how much work would be done?
- 27** If a charge of +3 C was placed at B and released
  - (a) what would happen to it?
  - (b) how much KE would it gain when it reached A?

## Potential and field strength

In the example of a uniform field the change in potential  $\Delta V$  when a charge is moved a distance  $\Delta h$  is given by

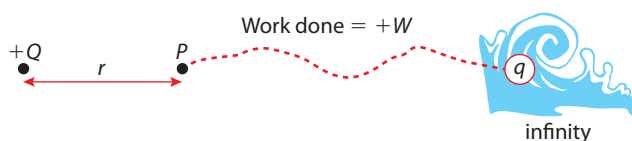
$$\Delta V = E\Delta h$$

Rearranging gives  $E = \frac{\Delta V}{\Delta h}$ , the field strength = the potential gradient. So in the example of Figure 6.33 the field strength is  $\frac{6}{3 \times 10^{-2}} \frac{\text{V}}{\text{m}} = 200 \text{ N C}^{-1}$

## Potential due to point charge

The uniform field is a rather special case; a more general example would be to consider the field due to a point charge. This would be particularly useful since all bodies are made of points; so if we know how to find the field due to one point we can find the field due to many points. In this way we can find the field caused by any charged object.

Consider a point P a distance  $r$  from a point charge  $Q$  as shown in Figure 6.34. The potential at P is defined as the work done per unit charge taking a small positive test charge from infinity to P.



### Contours

The lines of equipotential are again similar to contour lines but this time there is no real connection to gravity. Any hills and wells will be strictly imaginary.



### Permittivity

The constant  $k = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$

This can also be expressed in terms of the permittivity of a vacuum,  $\epsilon_0$

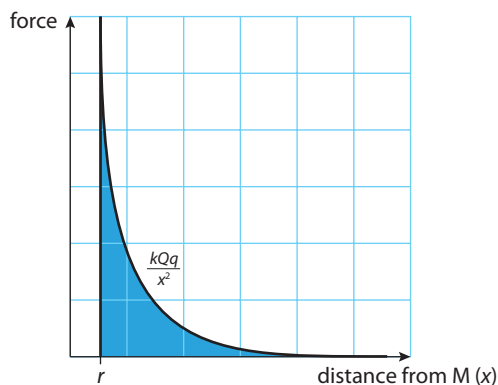
$$k = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

The permittivity is different for different media but we will only be concerned with fields in a vacuum.

**Figure 6.34** A positive charge is taken from infinity to point P.





**Figure 6.35** Graph of force against distance as charge  $+q$  approaches  $+Q$

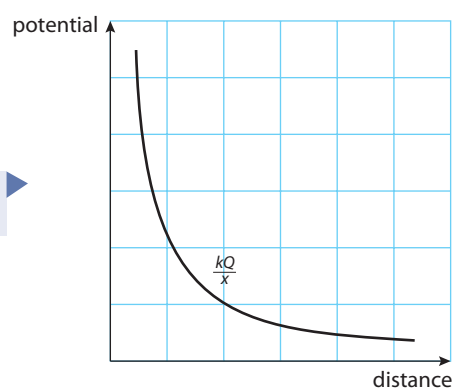
### Potential gradient

The potential gradient is related to the field strength by the equation

$$\frac{dV}{dx} = -E$$

We can see that this is negative because the gradient of  $V$  against  $x$  is negative but the field strength is positive.

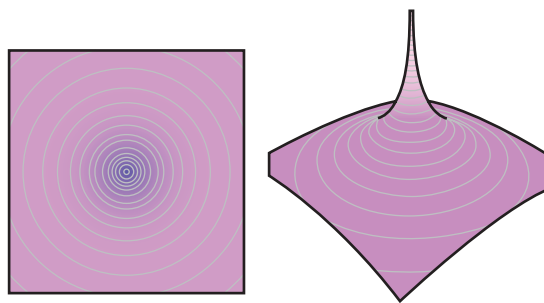
**Figure 6.36** Graph of potential against distance for a positive charge



## Equipotentials, wells and hills

If we draw lines of equipotential for a point charge we get concentric circles as shown in Figure 6.37. These look just the same as the gravitational equipotentials of Figure 6.15. However, we must remember that if the charge is positive, the potential increases as we get closer to it, rather than decreasing as in the case of gravitational field. These contours represent a hill not a well.

**Figure 6.37** The equipotentials and potential hill of a positive point charge.



### Exercises

- 28** Calculate the electric potential a distance 20 cm from the centre of a small sphere of charge  $+50 \mu\text{C}$ .
- 29** Calculate the p.d. between the point in question 28 and a second point 40 cm from the centre of the sphere.

## Addition of potential

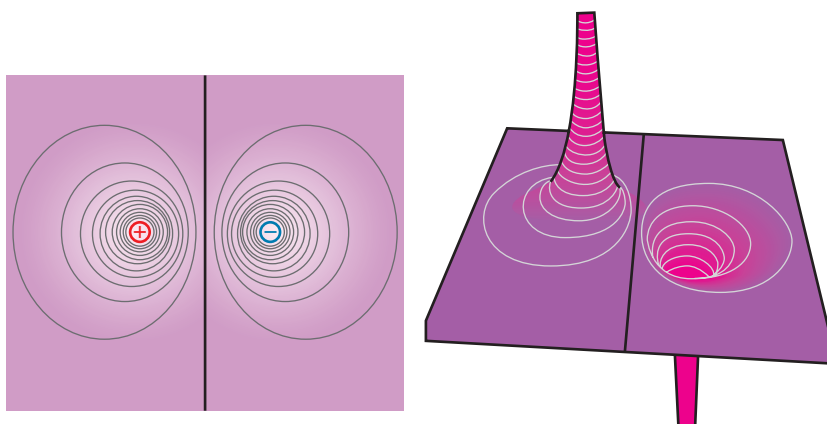
Since potential is a scalar there is no direction to worry about when adding the potential from different bodies – simply add them together.

### Example

At point P in Figure 6.38 the combined potential is given by:

$$V = k \frac{Q_1}{r_1} + k \frac{-Q_2}{r_2}$$

The potential of combinations of charging can be visualized by drawing lines of equipotentials. Figure 6.39 shows the equipotentials for a combination of a positive and negative charge (a dipole). This forms a hill and a well, and we can get a feeling for how a charge will behave in the field by imagining a ball rolling about on the surface shown.



● **Hint:** Zero potential is not zero field.

If we look at the hill and well in Figure 6.39, we can see that there is a position of zero potential in between the two charges (where the potentials cancel). This is not, however, a position of zero field since the fields will both point towards the right in between the charges.

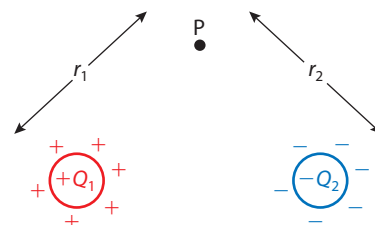
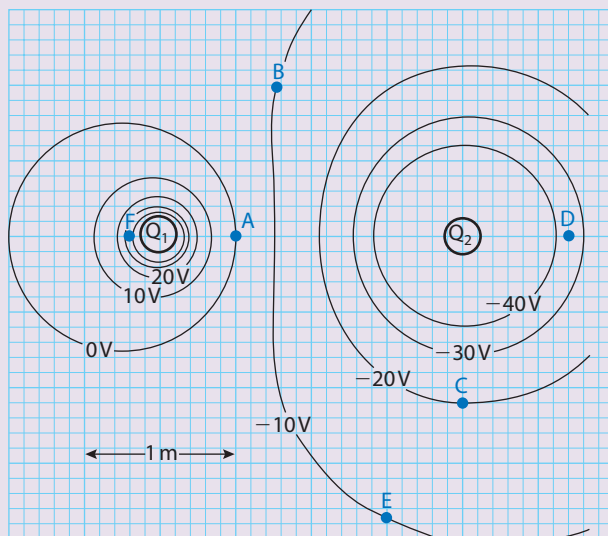


Figure 6.38

Figure 6.39 Equipotentials for a dipole.

## Exercises

Refer to Figure 6.40 for questions 30–36.



- 30 (a) One of the charges is positive and the other is negative. Which is which?  
 (b) If a positive charge were placed at A, would it move, and if so, in which direction?
- 31 At which point A, B, C, D or E is the field strength greatest?

Figure 6.40 The lines of equipotential drawn every 10V for two charges  $Q_1$  and  $Q_2$

**The electronvolt (eV)**

The electronvolt is a unit of electrical PE often used in atomic and nuclear physics. 1eV is the amount of energy gained by an electron accelerated through a p.d. of 1V.



- 32 What is the p.d. between the following pairs of points?  
 (a) A and C (b) C and E (c) B and E
- 33 How much work would be done taking a  $+2\text{ C}$  charge between the following points?  
 (a) C to A (b) E to C (c) B to E
- 34 Using the scale on the diagram, estimate the field strength at point D. Why is this an estimate?
- 35 Write an equation for the potential at point A due to  $Q_1$  and  $Q_2$ . If the charge  $Q_1$  is  $1\text{ nC}$ , find the value  $Q_2$ .
- 36 If an electron is moved between the following points, calculate the work done in eV (remember an electron is negative).  
 (a) E to A (b) C to F (c) A to C

## 6.7 Magnetic force and field

### Assessment statements

- 6.3.1 State that moving charges give rise to magnetic fields.
- 6.3.2 Draw magnetic field patterns due to currents.
- 6.3.3 Determine the direction of the force on a current-carrying conductor in a magnetic field.
- 6.3.4 Determine the direction of the force on a charge moving in a magnetic field.
- 6.3.5 Define the magnitude and direction of a magnetic field.
- 6.3.6 Solve problems involving magnetic forces, fields and currents.



Not all magnets are man-made; certain rocks (for example, this piece of magnetite) are naturally magnetic.

There is evidence in ancient Greek and Chinese writing, that people knew about magnets more than 2600 years ago. We do not know who discovered the first magnet, but since antiquity it was known that if a small piece of magnetite was suspended on a string and held close to another larger piece, the small one experienced a turning force, causing it to rotate. It was also found that if the small rock was held on its own, it would always turn to point toward the North Pole. The end of the rock that pointed north was named the *north-seeking pole*, the other end was named the *south-seeking pole*.



### What is a magnet?

We all know that magnets are the things that stick notes to fridge doors, but do we understand the forces that cause magnets to behave in this way?

### Magnetic poles



The north-seeking pole (red) always points to the north.

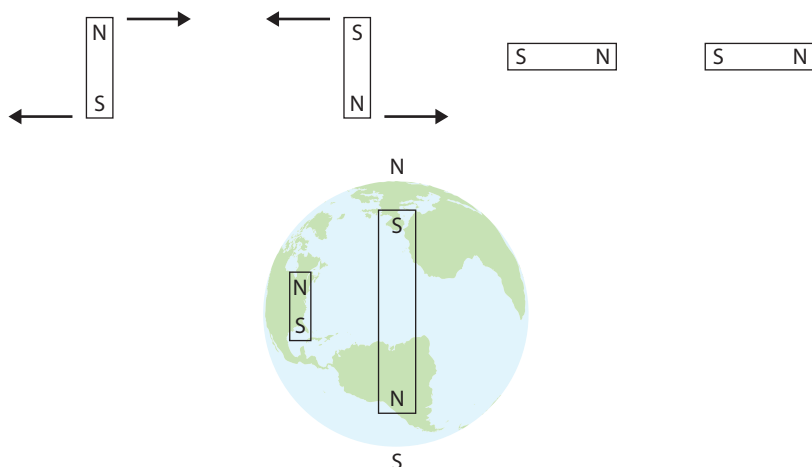
Every magnet has two poles (north and south). A magnet is therefore called a *dipole*. It is not possible to have a single magnetic pole or monopole. This is not the same as electricity where you can have a dipole or monopoles. If you cut a magnet in half, each half will have both poles.



**Figure 6.41** Magnets are dipoles.

## Unlike poles attract

If we take two magnets and hold them next to each other, we find that the magnets will turn so that the S and N poles come together.



**Figure 6.42** Magnets experience a turning force causing unlike poles to come together.

**Figure 6.43** The north-seeking pole of a compass points north.

We can therefore conclude that the reason that a small magnet points toward the North Pole of the Earth is because there is a south magnetic pole there. This can be a bit confusing, but remember that the proper name for the pole of the magnet is north-seeking pole.

## Magnetic field

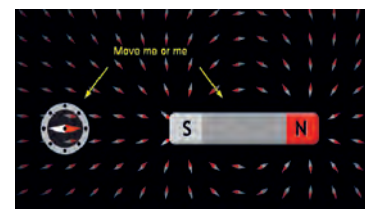
Magnetism is similar to gravitational force and electric force in that the effect is felt even though the magnets do not touch each other; we can therefore use the concept of field to model magnetism. However, magnetism isn't quite the same; we described both gravitational and electric fields in terms of the force experienced by a small mass or charge. A small magnet placed in a field does not accelerate – it rotates, and therefore magnetic field is defined as a region of space where a small test magnet experiences a turning force.



Since a small magnet rotates if held above the Earth, we can therefore conclude that the Earth has a magnetic field.

## Magnetic field lines

In practice, a small compass can be used as our test magnet. Magnetic field lines are drawn to show the direction that the N pole of a small compass would point if placed in the field.

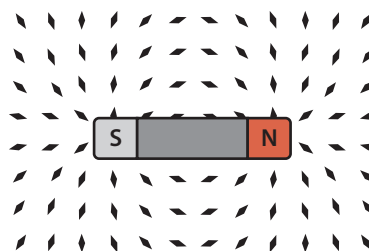


**W** To plot magnetic fields on the PhET 'Faraday's electromagnetic lab', visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on Weblink 6.5.

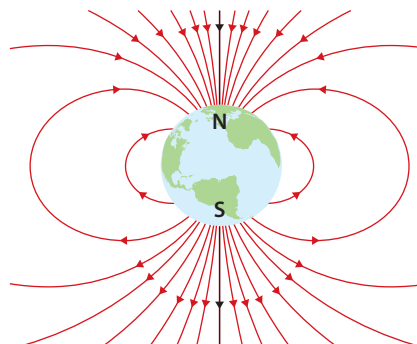
**Figure 6.44** The small magnet is caused to turn, so must be in a magnetic field.



**Figure 6.45** If the whole field were covered in small magnets, then they would show the direction of the field lines.



**Figure 6.46** The Earth's magnetic field.



### ***B* field**

Since the letter *B* is used to denote flux density, the magnetic field is often called a *B* field.

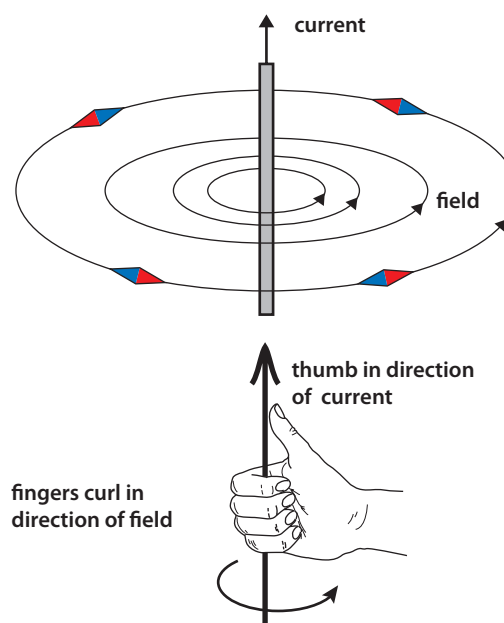
## **Magnetic flux density (*B*)**

From what we know about fields, the strength of a field is related to the density of field lines. This tells us that the magnetic field is strongest close to the poles. The magnetic flux density is the quantity that is used to measure how strong the field is – however it is not quite the same as field strength as used in gravitational and electric fields.

The unit of magnetic flux density is the tesla (T) and it is a vector quantity.

## **Field caused by currents**

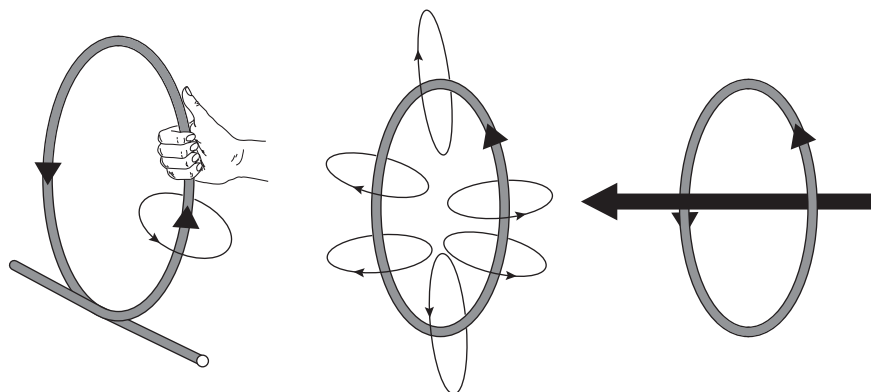
**Figure 6.47** The field due to a long straight wire carrying a current is in the form of concentric circles so the field is strongest close to the wire.



If a small compass is placed close to a straight wire carrying an electric current, then it experiences a turning force that makes it always point around the wire. The region around the wire is therefore a magnetic field. This leads us to believe that magnetic fields are caused by moving charges.

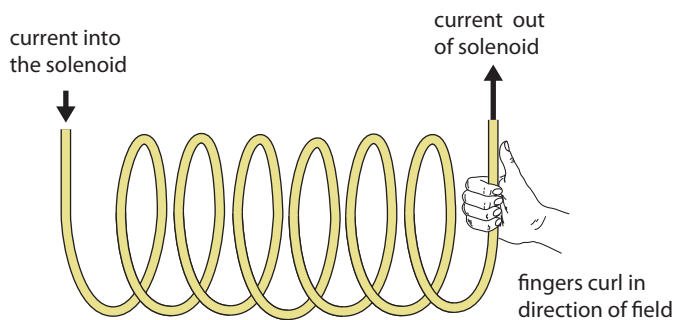
## Field inside a coil

When a current goes around a circular loop, the magnetic field forms circles.



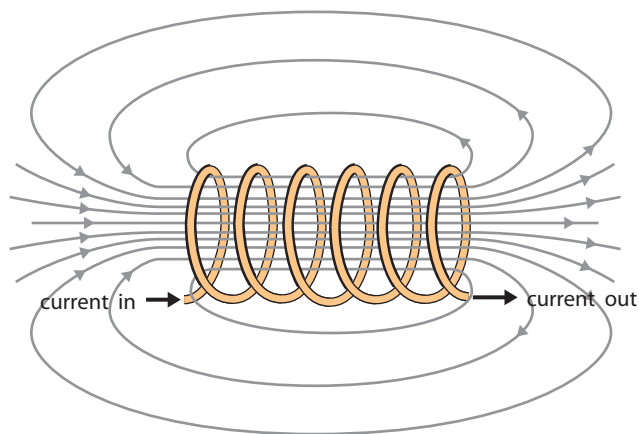
**Figure 6.48** The direction of the field can be found by applying the right-hand grip rule to the wire. The circles formed by each bit of the loop add together in the middle to give a stronger field.

## The field inside a solenoid



**Figure 6.49** The direction of the field in a solenoid can be found using the grip rule on one coil.

The resulting field pattern is like that of a bar magnet but the lines continue through the centre.

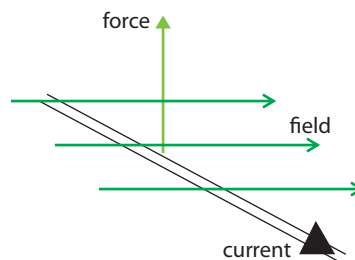


**Figure 6.50** The field inside a solenoid.

## Force on a current-carrying conductor

We have seen that when a small magnet is placed in a magnetic field, each end experiences a force that causes it to turn. If a straight wire is placed in a magnetic field, it also experiences a force. However, in the case of a wire, the direction of the force does not cause rotation – the force is in fact perpendicular to the direction of both current and field.

**Figure 6.51** Force, field and current are at right angles to each other.



### Definition of the ampere

The ampere is defined in terms of the force between two parallel current carrying conductors. A current of 1 A causes a force of  $2 \times 10^{-7}$  N per meter between two long parallel wires placed 1 m apart in a vacuum.



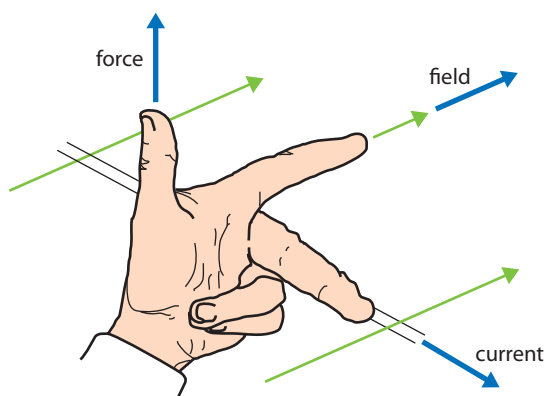
The size of this force is dependent on:

- how strong the field is – flux density  $B$
- how much current is flowing through the wire –  $I$
- the length of the wire –  $l$

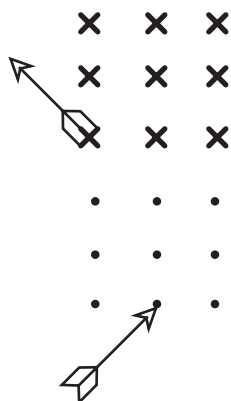
If  $B$  is measured in tesla,  $I$  in amps and  $l$  in metres,

$$F = BIl$$

**Figure 6.52** Using Fleming's left hand rule to find the direction of the force.



**Figure 6.53** Field into the page can be represented by crosses, and field out by dots. Think what it would be like looking at an arrow from the ends.



### Worked example

What is the force experienced by a 30 cm long straight wire carrying a 2 A current, placed in a perpendicular magnetic field of flux density  $6 \mu\text{T}$ ?

#### Solution

$$B = 6 \mu\text{T}$$

$$I = 2 \text{ A}$$

$$l = 0.3 \text{ m}$$

$$F = 6 \times 10^{-6} \times 2 \times 0.3 \mu\text{N}$$

$$= 3.6 \mu\text{N}$$

• **Examiner's hint:** Use the formula  $F = B \times I \times l$

### Exercises

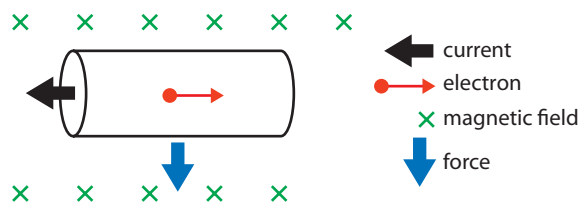
**37** A straight wire of length 0.5 m carries a current of 2 A in a north–south direction. If the wire is placed in a magnetic field of  $20 \mu\text{T}$  directed vertically downwards

- what is the size of the force on the wire?
- what is the direction of the force on the wire?

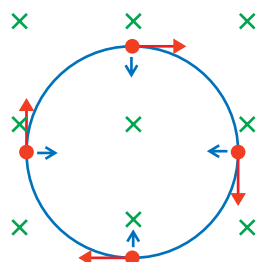
**38** A vertical wire of length 1 m carries a current of 0.5 A upwards. If the wire is placed in a magnetic field of strength  $10 \mu\text{T}$  directed towards the N geographic pole

- what is the size of the force on the wire?
- what is the direction of the force on the wire?

## Charges in magnetic fields



From the microscopic model of electrical current, we believe that the current is made up of charged particles (electrons) moving through the metal. Each electron experiences a force as it travels through the magnetic field; the sum of all these forces gives the total force on the wire. If a free charge moves through a magnetic field, then it will also experience a force. The direction of the force is always perpendicular to the direction of motion, and this results in a circular path.



**Figure 6.54** The force experienced by each electron is in the downward direction. Remember the electrons flow in the opposite direction to the conventional current.



The force on each charge  $q$  moving with velocity  $v$  perpendicular to a field  $B$  is given by the formula  $F = Bqv$ .

**Figure 6.55** Wherever you apply Fleming's left hand rule, the force is always towards the centre.

**Hint:** Remember – the electron is negative, so current is in opposite direction to electron flow.

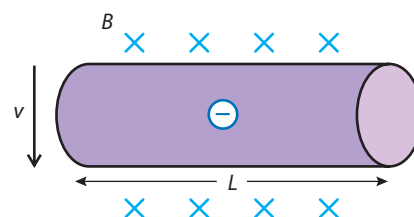
## 6.8 Electromagnetic induction

### Assessment statements

- 12.1.1 Describe the inducing of an emf by relative motion between a conductor and a magnetic field.
- 12.1.2 Derive the formula for the emf induced in a straight conductor moving in a magnetic field.
- 12.1.3 Define magnetic flux and magnetic flux linkage.
- 12.1.4 Describe the production of an induced emf by a time-changing magnetic flux.
- 12.1.5 State Faraday's law and Lenz's law.
- 12.1.6 Solve electromagnetic induction problems.

### Conductor moving in a magnetic field

We have considered what happens to free charges moving in a magnetic field, but what happens if these charges are contained in a conductor? Figure 6.56 shows a conductor of length  $L$  moving with velocity  $v$  through a perpendicular field of flux density  $B$ . We know from our microscopic model of conduction that conductors contain free electrons. As the free electron shown moves downwards through the field it will experience a force. Using Fleming's left hand rule, we can deduce that the direction of the force is to the left. (Remember, the electron is negative so if it is moving downwards the current is upwards.) This force will cause the free electrons to move to the left as shown in Figure 6.57. We can see that the electrons moving left have caused the lattice atoms on the right to become

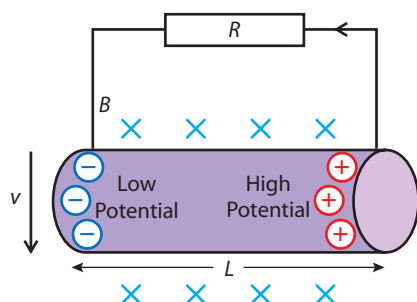


**Figure 6.56** A conductor moving through a perpendicular field.



positive, and there is now a potential difference between the ends of the conductor. The electrons will now stop moving because the  $B$  force pushing them left will be balanced by an  $E$  force pulling them right.

**Figure 6.57** Current flows from high potential to low potential.



## Induced emf

If we connect this moving conductor to a resistor, a current would flow, as the moving wire is behaving like a battery. If current flows through a resistor then energy will be released as heat. This energy must have come from the moving wire. When a battery is connected to a resistor, the energy released comes from the chemical energy in the battery. We defined the amount of chemical energy converted to electrical energy per unit charge as the emf. We use the same quantity here but call it the induced emf.

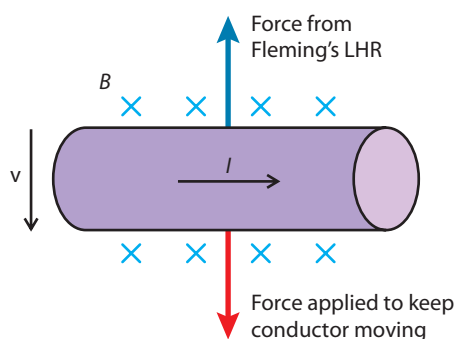
**The emf is the amount of mechanical energy converted into electrical energy per unit charge.**

The unit of induced emf is the volt.

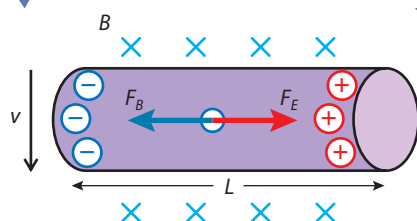
## Conservation of energy

When current flows through the resistor, current will also flow from left to right through the moving conductor. We now have a current carrying conductor in a magnetic field, allowing more electrons to move to the right. There is now a current flowing through the conductor from right to left. We now have a current-carrying conductor in a magnetic field, which, according to Fleming's left hand rule, will experience a force upwards (see Figure 6.58). This means that to keep it moving at constant velocity we must exert a force downwards, which means that we are now doing work. This work increases the electrical PE of the charges; when the charges then flow through the resistor this electrical PE is converted to heat, and energy is conserved.

**Figure 6.58** A constant force must be applied to the conductor to keep it moving.



**Figure 6.59** The electric and magnetic forces acting on the electron are balanced.



## Calculating induced emf

The maximum p.d. achieved across the conductor is when the magnetic force pushing the electrons left equals the electric force pushing them right. When the forces are balanced, no more electrons will move. Figure 6.59 shows an electron with balanced forces.



### The second law of thermodynamics

You can see that when the electrons are pushed to the left of the conductor they are becoming more ordered. So that this does not break the second law, there must be some disorder taking place.

If  $F_B$  is the magnetic force and  $F_E$  is the electric force we can say that

$$F_B = F_E$$

Now we know that if the velocity of the electron is  $v$  and the field strength is  $B$  then  $F_B = Bev$

The electric force is due to the electric field  $E$  which we can find from the equation  $E = -\frac{dV}{dx}$  that we established in the section on electric potential. In this case, the field is uniform so the potential gradient  $= \frac{V}{L}$

So, 
$$F_E = Ee = \frac{Ve}{L}$$

Equating the forces gives 
$$\frac{Ve}{L} = Bev$$

so 
$$V = BLv$$

This is the p.d. across the conductor, which is defined as the work done per unit charge, taking a small positive test charge from one side to the other. As current starts to flow in an external circuit, the work done by the pulling force enables charges to move from one end to the other, so the emf (mechanical energy converted to electrical per unit charge) is the same as this p.d.

$$\text{induced emf} = BLv$$

### Non-perpendicular field

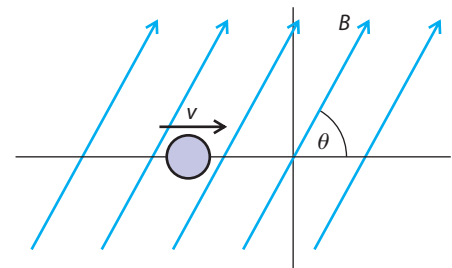
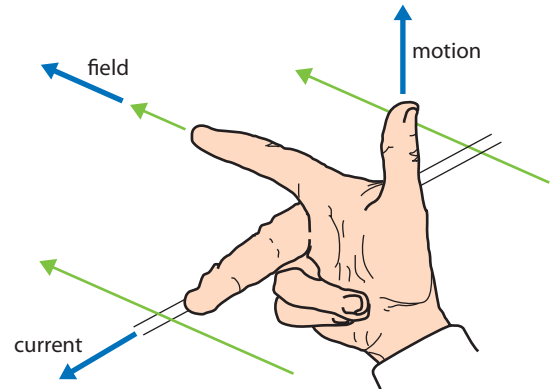
If the field is not perpendicular to the direction of motion then you take the component of the flux density that is perpendicular. In the example in Figure 6.60, this is  $B \sin \theta$ .

So  $\text{emf} = B \sin \theta \times Lv$

#### Hint:

##### Fleming's right hand rule

The fingers represent the same things as in the left hand rule but it is used to find the direction of induced current if you know the motion of the wire and the field. Try using it on the example in Figure 6.58.



**Figure 6.60** Looking along the wire, the component of the flux density that is perpendicular to the direction of motion is  $B \sin \theta$ .

### Exercises

- 39** A 20 cm long straight wire is travelling at a constant  $20 \text{ m s}^{-1}$  through a perpendicular B field of flux density  $50 \mu\text{T}$ .
- Calculate the emf induced.
  - If this wire were connected to a resistance of  $2 \Omega$  how much current would flow?
  - How much energy would be converted to heat in the resistor in 1s? (Power  $= I^2 R$ )
  - How much work would be done by the pulling force in 1s?
  - How far would the wire move in 1s?
  - What force would be applied to the wire?

## Faraday's law

From the moving conductor example we see that the induced emf is dependent on the flux density, speed of movement and length of conductor. These three factors all change the rate at which the conductor cuts through the field lines, so a more convenient way of expressing this is:

**The induced emf is equal to the rate of change of flux.**

This is Faraday's law and it applies to all examples of induced emf not just wires moving through fields.



This can be written as  $E = \frac{d\Phi}{dt}$

● **Hint:** If you think of the field lines as grass stems (they've been drawn green to help your imagination) and the conductor as a blade, then the induced emf is proportional to the rate of grass cut. This can be increased by moving the blade more quickly, having a longer blade or moving to somewhere where there is more grass.

## Flux and flux density

We can think of flux density as being proportional to the number of field lines per unit area, so flux is proportional to the number of field lines in a given area. If we take the example in Figure 6.61, the flux density of the field shown is  $B$  and the flux  $\varphi$  enclosed by the shaded area is  $BA$ .

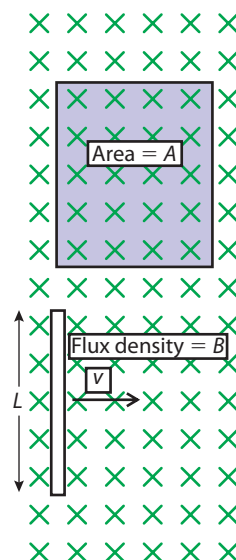
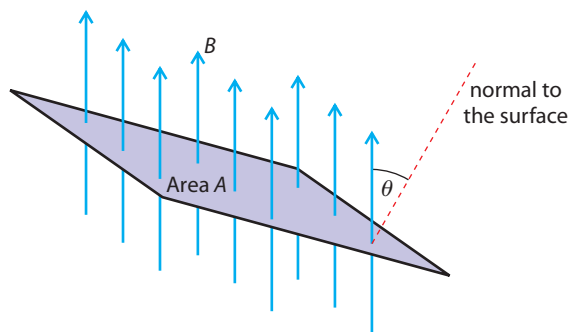
The unit of flux is tesla metre<sup>2</sup> (Tm<sup>2</sup>).

The wire in the same diagram will move a distance  $v$  in 1s (velocity is displacement per unit time) so the area swept out per unit time =  $Lv$ . The flux cut per unit time will therefore be  $BLv$ ; this is equal to the emf.

If the field is not perpendicular to the area then you use the component of the field that is perpendicular. In the example in Figure 6.62 this would give

$$\varphi = B \cos \theta \times A$$

**Figure 6.62** Area  $A$  not perpendicular to the field.



**Figure 6.61** A wire is moving through a uniform  $B$  field

## Lenz's law

We noticed that when a current is induced in a moving conductor, the direction of induced current causes the conductor to experience a force that opposes its motion. To keep the conductor moving will therefore require a force to be exerted in the opposite direction. This is a direct consequence of the law of conservation of energy. If it were not true, you wouldn't have to do work to move the conductor, so the energy given to the circuit would come from nowhere. Lenz's law states this fact in a way that is applicable to all examples:

**The direction of the induced current is such that it will oppose the change producing it.**

$$E = -\frac{dN\Phi}{dt}$$

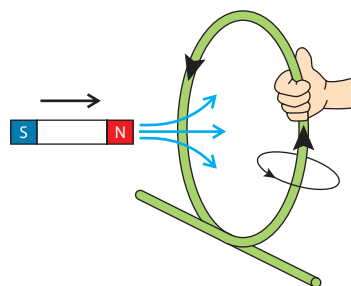
## Examples

### 1 Coil and magnet

A magnet is moved towards a coil as in Figure 6.63.

*Applying Faraday's law:*

As the magnet approaches the coil, the  $B$  field inside the coil increases, and the changing flux enclosed by the coil induces an emf in the coil that causes a current to flow. The size of the emf will be equal to the rate of change of flux enclosed by the coil.



**Figure 6.63** To oppose the magnet coming into the coil, the coil's magnetic field must push it out. The direction of the current is found using the grip rule.

### Applying Lenz's law:

The direction of induced current will be such that it opposes the change producing it, which in this case is the magnet moving towards the coil. So to oppose this, the current in the coil must induce a magnetic field that pushes the magnet away; this direction is shown in the diagram.

## 2 Coil in a changing field

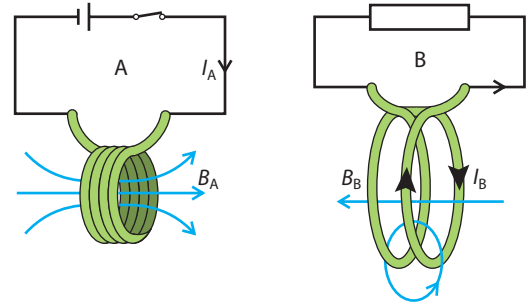
In Figure 6.64, the magnetic flux enclosed by coil B is changed by switching the current in coil A on and off.

### Applying Faraday's law:

When the current in A flows, a magnetic field is created that causes the magnetic flux enclosed by B to increase. This increasing flux induces a current in coil B.

### Applying Lenz's law:

The direction of the current in B must oppose the change producing it, which in this case is the increasing field from A. So to oppose this, the field induced in B must be in the opposite direction to the field from A, as in the diagram. This is the principle behind the operation of a transformer.



**Figure 6.64** Coil A induces current in coil B. Use the grip rule to work out the direction of the fields.

## Exercises

- 40** A coil with 50 turns and area  $2 \text{ cm}^2$  encloses a field that is of flux density  $100 \mu\text{T}$  (the field is perpendicular to the plane of the coil).
- What is the total flux enclosed?
  - If the flux density changes to  $50 \mu\text{T}$  in 2 s, what is the rate of change of flux?
  - What is the induced emf?
- 41** A rectangular coil with sides 3 cm and 2 cm and 50 turns lies flat on a table in a region of magnetic field. The magnetic field is vertical and has flux density  $500 \mu\text{T}$ .
- What is the total flux enclosed by the coil?
  - If one side of the coil is lifted so that the plane of the coil makes an angle of  $30^\circ$  to the table, what will the new flux enclosed be?
  - If the coil is lifted in 3 s, estimate the emf induced in the coil.

● **Hint:** If a coil has  $N$  turns each turn of the wire encloses the flux. If the flux enclosed by the area of the coil is  $\phi$  then the total flux enclosed is  $N\phi$ .

## Applications of electromagnetic induction

### Induction braking

Traditional car brakes use friction pads that press against a disc attached to the road wheel. Induction braking systems replace the friction pads with electromagnets. When switched on, a current is induced in the rotating discs. According to Lenz's law the induced current will oppose the change producing it, resulting in a force that slows down the car.

### Induction cooking

An induction hotplate uses a changing magnetic field to induce an emf in a metal saucepan; the emf causes a current to flow in the saucepan, which produces heat ( $I^2R$ ). The benefit of this system is that the saucepan (and therefore the food) gets hot but not the oven; you can touch the 'hotplate' without getting burned.



In an induction braking system there are no parts to wear out, which means you would not have to replace brake pads every couple of years. This is bad news for the manufacturers of brake pads. Sometimes advancements in technology are not developed because they do not benefit manufacturers.

## 6.9 Alternating current

### Assessment statements

- 12.2.1 Describe the emf induced in a coil rotating within a uniform magnetic field.
- 12.2.2 Explain the operation of a basic alternating current (AC) generator.
- 12.2.3 Describe the effect on the induced emf of changing the generator frequency.
- 12.2.4 Discuss what is meant by the root mean squared (rms) value of an alternating current or voltage.
- 12.2.5 State the relation between peak and rms values for sinusoidal currents and voltages.
- 12.2.6 Solve problems using peak and rms values.
- 12.2.7 Solve AC circuit problems for ohmic resistors.
- 12.2.8 Describe the operation of an ideal transformer.
- 12.2.9 Solve problems on the operation of ideal transformers.

To view a simulation of an electric generator, visit [www.heinemann.co.uk/hotlinks](http://www.heinemann.co.uk/hotlinks), enter the express code 4426P and click on Weblink 6.6.

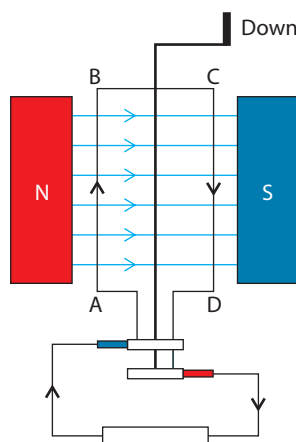


The most common application of electromagnetic induction is in the production of electrical energy. There are several devices that can be used to do this, such as the dynamo, in which a coil rotates in a magnetic field, or an alternator, in which a magnet rotates in a coil. Here we will consider the simple case of a coil rotating in a uniform magnetic field.

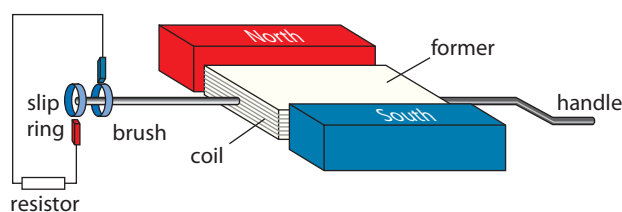
### Coil rotating in a uniform magnetic field

Consider the coil shown in Figure 6.65. This coil is being made to rotate in a uniform magnetic field by someone turning the handle. The coil is connected to a resistor, but to prevent the wires connected to the coil twisting, they are connected via two slip rings. Resting on each slip ring is a carbon brush, which makes contact with the ring whilst allowing it to slip past.

**Figure 6.65** A simple AC generator.



**Figure 6.66** Looking at the generator from above.



To make the operation easier to understand, a simpler 2D version with only one loop of wire in the coil is shown in Figure 6.66. As the handle is turned, the wire on the left hand side (AB) moves up through the field. As it cuts the field a current will be induced. Using Fleming's right hand rule, we can deduce that the direction of the current is from A to B as shown. The direction of motion of the right hand side (CD) is opposite so the current is opposite. The result is a clockwise current through the resistor.

After turning half a revolution the coil is in the position shown in Figure 6.67. Side CD is now moving up through the field. Look carefully at how the slip ring has moved and you will see why, although the current is still clockwise in the coil, it is anticlockwise in the resistor circuit.

## The size of the emf induced in a rotating coil

To find the size of the emf, we can use Faraday's law. This states that the induced emf will be equal to the rate of change of flux. The flux enclosed by the coil is related to the angle the coil makes with the field. Figure 6.68 shows a coil of  $N$  turns at time  $t$ . At this moment the normal to the plane of the coil makes an angle  $\theta$  with the field.

$$\text{Flux } \varphi = BA \cos \theta$$

There are  $N$  turns so total flux  $N\varphi = BAN \cos \theta$  (1)

If the angular velocity of the coil is  $\omega$  then the angle  $\theta = \omega t$

Substituting equation (1) gives

$$N\varphi = BAN \cos \omega t$$

This equation can be represented graphically as in Figure 6.69.

Note that the graph starts when the flux enclosed is a maximum, so  $\cos \theta = 1$  meaning that  $\theta = 0^\circ$ . Remember that  $\theta$  is the angle between the field and the normal to the plane of the coil, so position A is when the coil in Figure 6.68 is vertical.

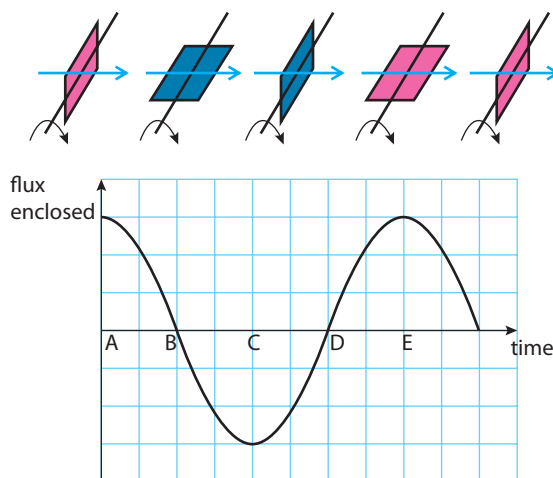
To find the magnitude of the emf, we need the rate of change of flux. We can calculate this from the gradient of this graph. Let us consider some specific points.

At A the gradient is zero.

At B the gradient is big and negative.

At C the gradient is zero.

At D the gradient is big and positive.

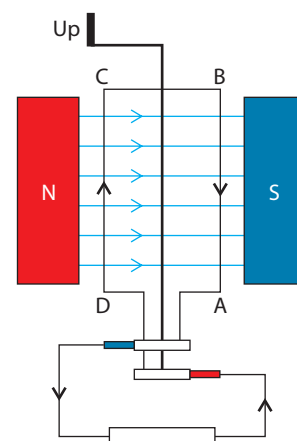


Lenz's law says that the induced current will oppose the change producing it; this means that when the rate change of flux is positive the induced emf is negative. If we plot the rate of change of flux against time we get a graph as in Figure 6.70. The equation of this line is

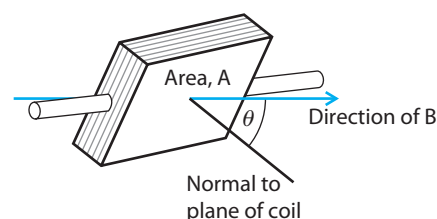
$$E = BAN\omega \sin \omega t$$

Where  $BAN\omega$  is the peak value  $E_0$

Note that this is a maximum when the flux enclosed is zero, that is when the coil is in the position shown in Figure 6.65.



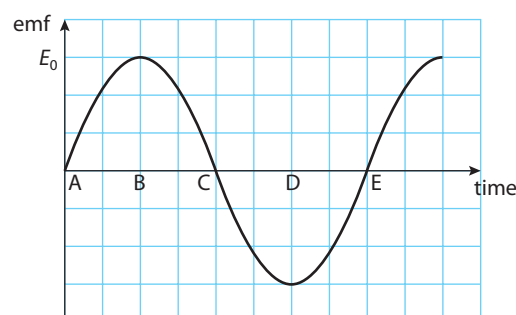
**Figure 6.67** The coil after half a revolution.



**Figure 6.68** Coil at an angle to the field.

**Figure 6.69** Graph of flux against time.

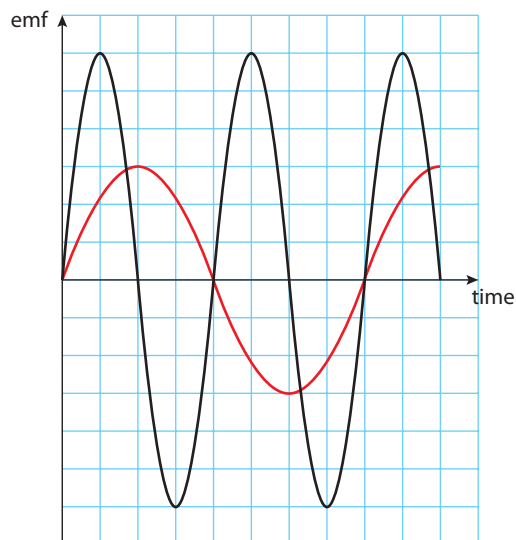
**Figure 6.70** Graph of emf against time.



### Effect of increasing angular speed

If the speed of rotation is increased, the graph of emf against time will change in two ways, as shown in Figure 6.71. Firstly, time between the peaks will be shorter, and secondly, the peaks will be higher. This is because if the coil moves faster, then the rate of change of flux will be higher and hence the emf will be greater.

**Figure 6.71** The black graph is for a coil with twice the angular speed of the red one.



#### Faraday's law

If you have studied differentiation in maths you will understand that Faraday's law can be written

$$\epsilon = \frac{-dN\phi}{dt}$$

Then if  $N\phi = BAN \cos \omega t$

$$\frac{-dN\phi}{dt} = BAN\omega \sin \omega t$$

### Alternating current

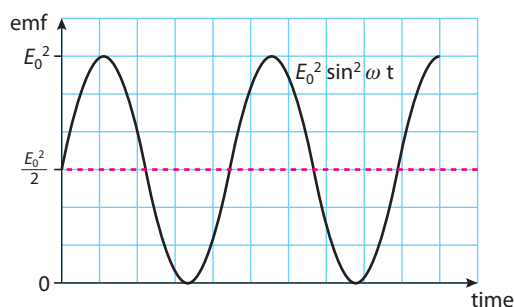
The current delivered by the rotating coil changes in direction and size over a period of time. This is called *alternating current* (AC). A battery, however, gives a constant current called *direct current* (DC). When doing power calculations with AC (for example, calculating the heat given out per second by an electric heater), the peak value would give a result that was too big since the peak value is only attained for a very short time. In these cases we should use the root mean square value; this is a sort of average value.

#### Root mean square

The root mean square or rms is the square root of the mean of the squares. Since the emf from the rotating coil varies sinusoidally then the rms emf and current will be the same as the root of the mean of the squares of the sine function.

To calculate the rms value, first we must square the function; this gives the curve shown in Figure 6.72. If we consider one complete cycle of this function we see that the mean value is  $\frac{1}{2} E_0^2$ . The rms value is then the square root of this.

**Figure 6.72**



$$E_{\text{rms}} = \sqrt{\frac{E_0^2}{2}} = \frac{E_0}{\sqrt{2}}$$

The current passing through a resistor will be proportional to the potential difference across it, so this will also be sinusoidal. If the peak current is  $I_0$  the rms value will therefore be given by:

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

The rms values are what you need to know for power calculations, so it is these that are normally quoted. For example, the rms mains voltage in Europe = 220 V.

## Power in AC circuits

If the rms current flowing through a resistor is  $I_{\text{rms}}$  and the rms potential difference across it is  $V_{\text{rms}}$  then the power dissipated in it will be  $I_{\text{rms}} \times V_{\text{rms}}$ .

### Worked example

What will be the rms current flowing through a 100 W light bulb connected to a 220 V AC supply?

#### Solution

$$\text{Power} = I_{\text{rms}} V_{\text{rms}}$$

$$\text{So } I_{\text{rms}} = \frac{\text{Power}}{V_{\text{rms}}} = \frac{100}{220} = 0.45 \text{ A}$$



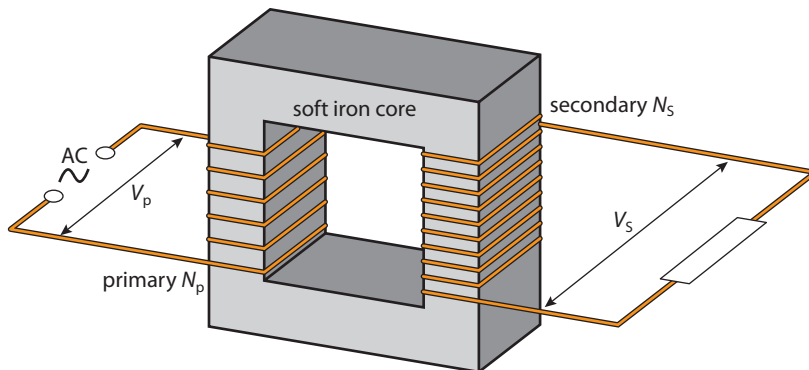
The electrical energy produced by a generator comes from the work done by the person turning the coil. The more current you take from the coil the harder it is to turn. This follows from Lenz's law; the current in the coil opposes the change producing it. If you don't draw any current then it's very easy to turn the coil.

### Exercises

- 42 The rms voltage in the USA is 110V. Calculate the peak voltage.
- 43 An electric oven designed to operate at 220V has a power rating of 4 kW. What current flows through it when it is switched on?
- 44 A coil similar to the one in Figure 6.65 has an area of  $5 \text{ cm}^2$  and rotates 50 times a second in a field of flux density 50 mT.
- If the coil has 500 turns, calculate
    - the angular velocity,  $\omega$
    - the maximum induced emf
    - the rms emf.
  - If the speed is reduced to 25 revolutions per second what is the new  $E_{\text{rms}}$ ?
- 45 Calculate the resistance of a 1000W light bulb designed to operate at 220V.

## The transformer

A transformer consists of two coils wound on a common soft iron core as shown in Figure 6.73. The primary coil is connected to an AC supply. This causes a changing magnetic field inside the coil. This field is made stronger by the presence of the soft iron core which itself becomes magnetic (temporarily). Since the secondary is wound around the same former, it will have a changing magnetic field within it, which induces an emf in its coils.



The emf induced in the secondary is directly related to the number of turns. If the supply is sinusoidal then the ratio of turns in the two coils equals the ratio of the p.d.s.

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

Figure 6.73 A simple transformer.



#### Power losses

Real transformers are not 100% efficient; power is lost in several ways:

- Heat in the wire of the coils
- Heat in the core due to induced currents in the soft iron.
- Flux leakage.
- The core retains some magnetism when the current changes.



So a transformer can make a p.d. higher or lower depending on the ratio of turns on the primary and secondary. However, since energy must be conserved, the power out cannot be bigger than the power in. Electrical power is given by  $I \times V$  so if the p.d. goes up, the current must come down. An ideal transformer has an efficiency of 100%. This means that power in = power out.

$$V_p I_p = V_s I_s$$

where  $V_p$ ,  $V_s$ ,  $I_p$  and  $I_s$  are the rms values of p.d. and current in the primary and secondary.

### Warning



The emf induced in the secondary depends on the rate of change of current in the primary. If you switch the current off then the change can be very big inducing a much bigger emf than you might have calculated.

### Exercises

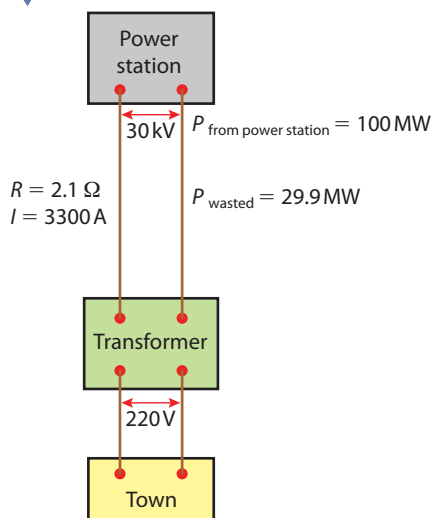
- 46** An ideal transformer steps down the 220V mains to 4.5V so it can be used to charge a mobile phone.
- If the primary has 500 turns how many turns does the secondary have?
  - If the charger delivers 0.45 A to the phone, how much power does it deliver?
  - How much current flows into the charger from the mains?
  - The phone is unplugged from the charger but the charger is left plugged in. How much current flows into the charger now?

## 6.10 Transmission of electrical power

### Assessment statements

- 12.3.1 Outline the reasons for power losses in transmission lines and real transformers.
- 12.3.2 Explain the use of high-voltage step-up and step-down transformers in the transmission of electrical power.
- 12.3.3 Solve problems on the operation of real transformers and power transmission.
- 12.3.4 Suggest how extra-low-frequency electromagnetic fields, such as those created by electrical appliances and power lines, induce currents within a human body.
- 12.3.5 Discuss some of the possible risks involved in living and working near high-voltage power lines.

Figure 6.74



As outlined previously, electrical energy can be produced by moving a coil in a magnetic field. The energy required to rotate the coil can come from many sources, for example: coal, oil, falling water, sunlight, waves in the sea or nuclear fuel. The transformation of energy takes place in power stations that are often not sited close to the places where people live. For that reason, the electrical energy must be delivered via cables. These cables have resistance so some energy will be lost as the current flows through them. Let's take an example as illustrated in Figure 6.74.

The generators at a power station typically produce 100 MW of power at 30 kV. A small town 100 km from the power station requires electrical power which will be delivered via two aluminium cables (one there and one back) that have a radius of 2 cm. The houses in the town cannot use such high voltage electricity so the p.d. must be stepped down to 220 V, using a transformer that we will assume is 100% efficient.

The first thing to do is to calculate the resistance of the cables. We can do this using the formula

$$R = \frac{\rho l}{A}$$

where  $\rho$  is the resistivity of aluminium ( $2.6 \times 10^{-8} \Omega\text{m}$ ),  $A$  is the cross-sectional area and  $l$  is the length.

$$R = \frac{2.6 \times 10^{-8} \times 200 \times 10^3}{2\pi \times (2 \times 10^{-2})^2} = 2.1 \Omega$$

The current that must be delivered if 100 MW is produced at a p.d. of 30 kV can be found using  $P = IV$

$$I = \frac{1 \times 10^8}{30 \times 10^3} = 3.3 \times 10^3 \text{ A}$$

The power loss in the cables is therefore  $I^2R = (3.3 \times 10^3)^2 \times 2.1 = 22.9 \text{ MW}$

This is a lot of wasted power.

To reduce the power loss in the cables we must reduce the current. This can be done by stepping up the voltage before transmission, using a transformer as shown in the next example (Figure 6.75).

The p.d. between the wires is typically stepped up to 115 kV. Let us now repeat the calculation, assuming all transformers are 100% efficient.

The power is now delivered at 115 kV so the current is

$$I = \frac{P}{V} = \frac{100 \times 10^6}{115 \times 10^3} = 870 \text{ A}$$

Power loss in the cables =  $I^2R = 870^2 \times 2.1 = 1.6 \text{ MW}$

This is still quite a lot of wasted power but much less than before.

To reduce this further, more cables could be added in parallel, thereby reducing the resistance.

## Health risks and power lines

Power lines carry large alternating currents at high potentials which produce electromagnetic fields radiating from them. The changing electric and magnetic field will induce currents in any conductors placed nearby. Your body is a conductor, so if you stand near a power line small currents will be induced in your body – is this harmful?

Unless you almost sit on the cable, these fields are very small. Typically the magnetic field you'd experience is less than the Earth's magnetic field. The frequency of the change is also very low so cannot affect the atoms of your body. It therefore seems very unlikely that any harm could be caused by these fields. In the 1970s, a study of the incidence of childhood leukaemia showed that there was a higher incidence of leukaemia in children living close to power lines. This caused a lot of concern at the time, but the study was seriously flawed. One of the main pieces of evidence against this theory is that even though the number of children living near power lines has increased significantly over the past 30 years, the incidence of leukaemia has gone down.

● **Hint:** 30 kV is the p.d. between the two wires not the p.d. from the power station to the transformer.

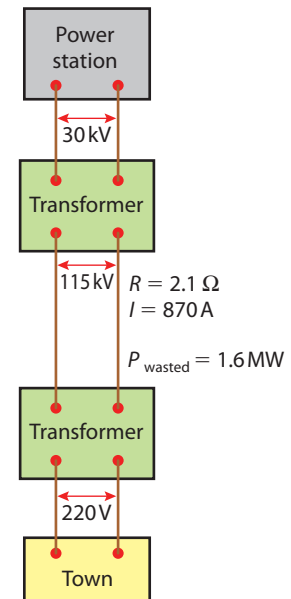


Figure 6.75



### Power lines and leukaemia

This is a good example of how people can be influenced by the media. Even though there is no scientific evidence that power lines adversely affect one's health, people still believe what they heard on the TV 30 years ago. There are also a lot of companies marketing products that profit from this misconception.